EclipsePhoto

Automated Photography of Total Solar Eclipses

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EclipsePhoto Project

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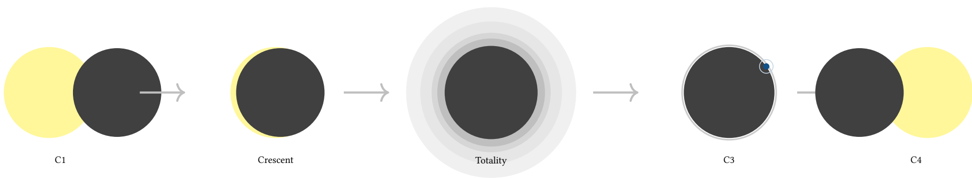

Abstract. EclipsePhoto is an autonomous eclipse photography control system designed to capture total solar eclipse events across their full dynamic and temporal range. Built on Raspberry Pi capture nodes coordinated by an iPhone hub, the system manages the critical challenges unique to eclipse photography: a 17 EV illuminance swing in minutes, safety-critical solar filter management, HDR bracket sequences spanning 12+ EV stops of corona brightness, and sub-second transient phenomena (diamond ring, chromosphere flash). This report details the astronomical science, optical engineering, and operational procedures that inform the system design, from eclipse geometry and contact-point timing through multi-rig deployment strategy and real-time telemetry.

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1 Problem Framing

Total solar eclipse photography presents a uniquely adversarial control problem. Unlike aurora photography—where displays recur and the photographer can iterate over multiple nights—a total eclipse at a given location is a once-in-a-lifetime event. The path of totality is roughly 150 km wide and sweeps across the Earth's surface at over 1 500 km/h; at any given site, totality lasts between 2 and 7 minutes. There are no second chances.

Within those few minutes, the photographer must contend with five interlocking challenges:

1. **Extreme illuminance range.** Ambient light drops from ~100,000 lux (full sunlight) to ~1 lux (deep twilight equivalent) in the final minutes before C2, a span of approximately 17 EV stops. The exposure ramp must track this change on a cadence of seconds, not minutes.
2. **Safety-critical solar filter management.** A solar ND filter (optical density ~5, transmitting ~0.001% of incident light) must be in place during all partial phases to protect both the camera sensor and the operator's eyes. Forgetting to *remove* the filter at C2 wastes the entire totality window. Forgetting to *replace* it at C3 risks immediate sensor damage from unfiltered sunlight.
3. **Sub-second transient phenomena.** The diamond ring effect (a single brilliant point of sunlight shining through a lunar valley) lasts 1–3 seconds. The chromosphere flash (a thin red ring of hydrogen emission) is visible for 2–5 seconds at C2 and again at C3. Both require pre-armed burst capture at the correct exposure—there is no time to meter and adjust.
4. **Multi-rig coordination.** A serious eclipse deployment involves two or three camera rigs at different focal lengths (wide-angle for context, telephoto for corona, long telephoto for prominences). Each rig requires synchronized filter management, exposure ramping, and bracket sequencing.
5. **Operator cognitive overload.** Totality triggers a well-documented "awe response" that degrades the operator's ability to perform precise manual tasks. Experienced eclipse photographers report forgetting entire steps of their carefully rehearsed plan. Automation is not merely convenient—it is necessary for reliable capture.

The remainder of this report develops the science and engineering behind each of these challenges. Sections 2–6 establish the astronomical context: eclipse geometry, illuminance curves, physical phenomena at each contact point, coronal dynamic range, and color temperature shifts. Section 7 maps these phenomena to optical parameters. Sections 8–10 translate the science into system design: exposure strategy, filter management, and multi-rig deployment. Sections 11–13 detail the operational pipeline, and Sections 14–16 address monitoring and robustness.

Insight: *The key product value of EclipsePhoto is not intervalometer-style automation but rather time-critical orchestration: managing filter state, exposure ramps, and bracket sequences across multiple rigs during a window where human cognition is unreliable.*

2 Eclipse Geometry and Contact Points

A total solar eclipse progresses through four contact points (C1–C4) as the Moon traverses the solar disc. The system must track these stages because illuminance changes by a factor of ~10,000× between partial phases and totality, requiring radically different exposure parameters at each transition.

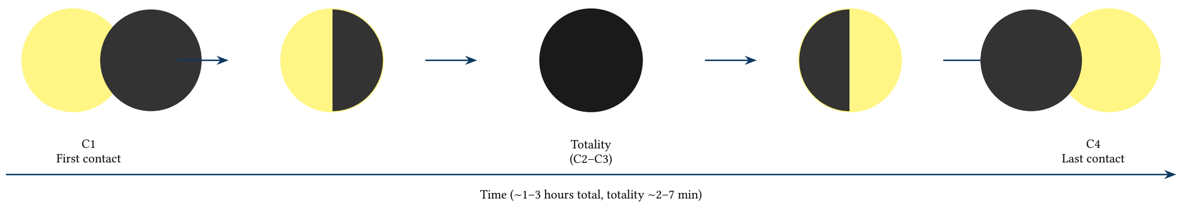


Figure 1. Solar eclipse progression: the five major phases from first contact (C1) through partial eclipse, totality, partial eclipse, to last contact (C4). The entire event spans 1–3 hours, with totality lasting 2–7 minutes.

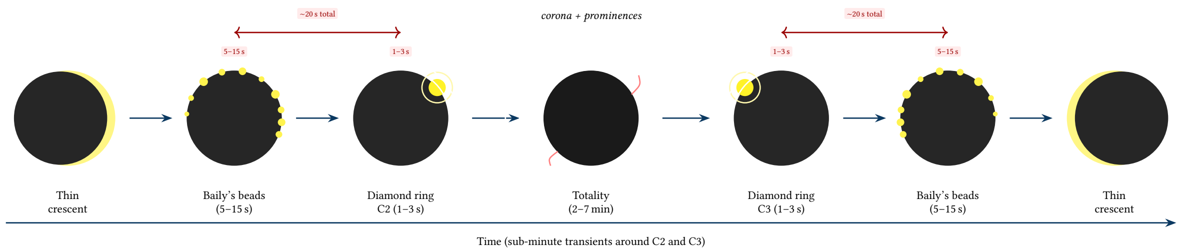


Figure 2. Detailed transition sequence at C2 and C3: the critical seconds surrounding second and third contact. As the last sliver of photosphere disappears, sunlight fragments into Baily's beads (multiple points through lunar valleys), then coalesces into a single brilliant diamond ring before totality. The reverse occurs at C3. These sub-minute transients require pre-armed burst capture.

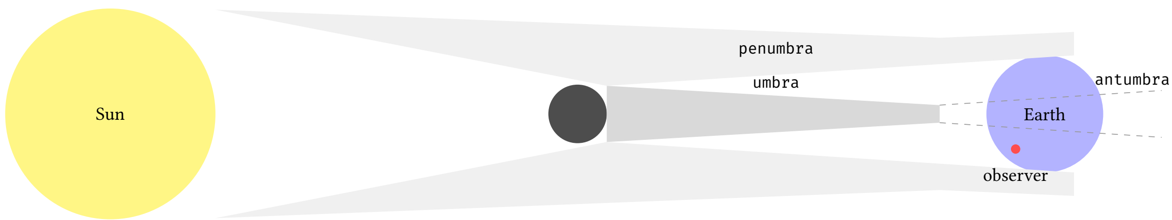


Figure 3. Eclipse shadow geometry (cross-section): the Moon casts an umbral cone (total shadow), penumbral cone (partial shadow), and antumbra (annular shadow region beyond the umbral vertex).

The shadow geometry directly determines what each observer sees. An observer in the penumbral cone sees a partial eclipse—the Moon covers part of the solar disc. An observer at the umbral cone's intersection with the Earth sees totality—the Moon fully occults the photosphere and the corona becomes visible. The transition from deep partial eclipse to totality occurs over a distance of just a few kilometres at the umbral boundary, which is why precise site selection along the centre line is critical.

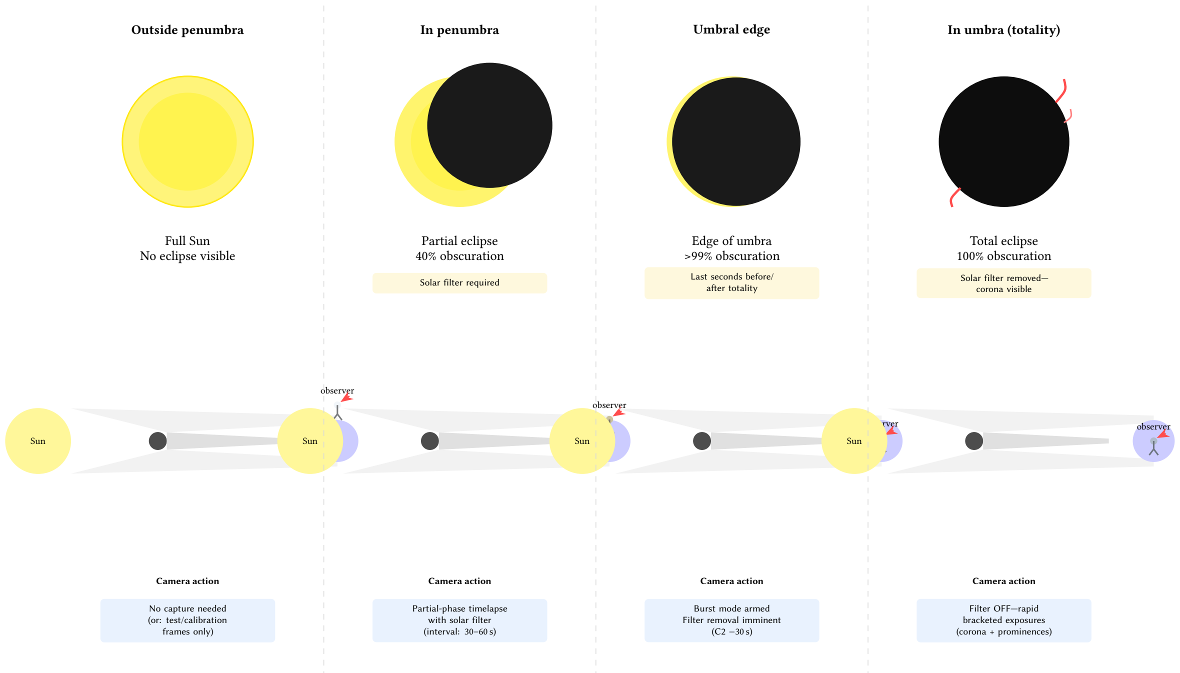


Figure 4. Observer's view at each shadow position: connecting the cross-section geometry (Figure above) to the visual experience and camera requirements. Each column shows the Sun/Moon appearance, the observer's location in the shadow cone, and the corresponding camera action.

The Moon's shadow traces a narrow path—the *path of totality*—across the Earth's surface as the Moon orbits and the Earth rotates. This totality band is typically 100–300 km wide, depending on the Moon's distance and the eclipse geometry. The shadow moves at 1,500–3,000 km/h, faster near the edges of the Earth's disc (oblique incidence) and slower near the sub-solar point. Partial eclipse is visible over a much wider area, extending thousands of kilometres on either side of the totality path. Despite the width of partial visibility, total eclipse at any given fixed point on the Earth's surface is rare—occurring once every 375 years on average. For eclipse photography, site selection along or near the centre line of the totality path maximises the duration of totality and therefore the available capture window.

Each contact point has specific photographic implications:

- **C1 (First Contact):** The Moon's limb first touches the solar disc. Begin partial-phase capture with the solar filter in place. A cadence of 1 frame every 2 minutes is sufficient to document the Moon's progression across the Sun. Sunspot transit can be captured if active regions are present.
- **C2 (Second Contact):** The Moon fully covers the solar disc and totality begins. This is the single most critical transition: the solar filter *must* be removed, and HDR bracket capture must begin immediately. The diamond ring effect occurs in the final 1–3 seconds before C2, requiring burst capture at fast shutter speeds (1/2000–1/4000s). The chromosphere flash (thin red Hα ring) is visible for 2–5 seconds around C2.
- **C3 (Third Contact):** Totality ends as the first sliver of photosphere reappears. The diamond ring and chromosphere flash occur again in the reverse sequence. Immediately after capturing the C3 diamond ring, the solar filter *must* be replaced to protect the sensor from the rapidly increasing sunlight.
- **C4 (Fourth Contact):** The Moon's limb fully separates from the solar disc. End of partial-phase capture. Stop the intervalometer sequence.

: Contact-Point State Mapping

Each contact point maps directly to a state-machine transition (Section 13): C1 triggers the partial eclipse state, C2 triggers filter removal, C3 triggers filter replacement, and C4 triggers completion. The system must receive accurate GPS-synchronized contact times to drive these transitions.

3 Light Levels Across Eclipse Stages

The illuminance change during a total solar eclipse spans roughly five orders of magnitude. Understanding this curve is essential for designing the exposure ramp strategy: the system must transition smoothly from bright-sun settings to near-nighttime settings and back within minutes.

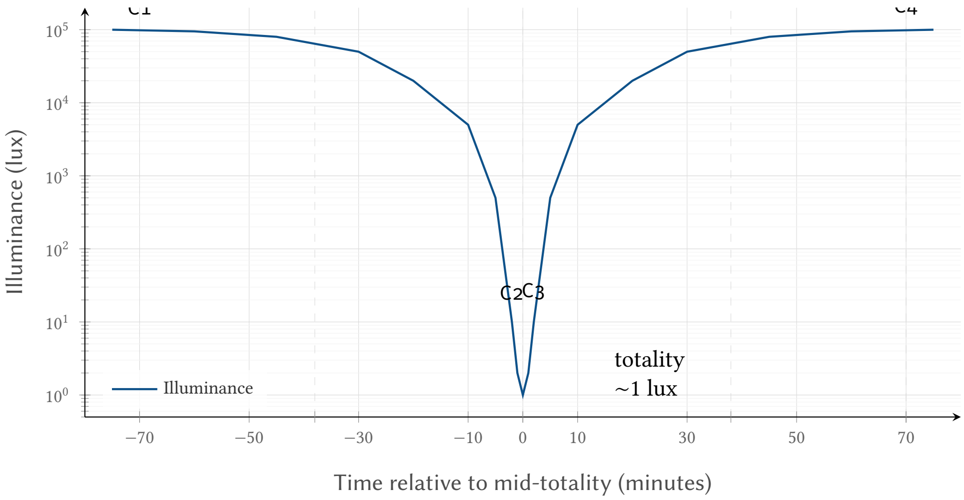


Figure 5. Solar eclipse illuminance curve (log scale). The drop from ~100,000 lux to ~1 lux occurs over just a few minutes around C2/C3, requiring a ~17 EV stop adjustment.

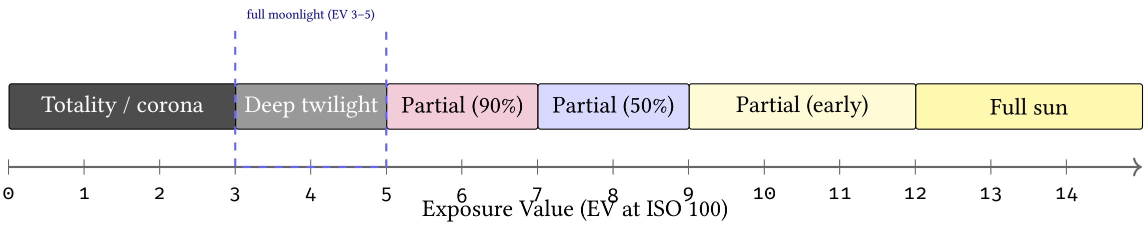


Figure 6. Equivalent EV scale showing the range of scene brightness from totality (EV 0–3) through partial eclipse stages to full sunlight (EV 14–15).

Eclipse Stage	Illuminance	EV	Notes
Full sun (pre-C1)	100,000 lux	14–15	Solar filter required
Partial (25% covered)	75,000 lux	14	Solar filter still required
Partial (50% covered)	50,000 lux	13	Solar filter still required
Partial (90% covered)	10,000 lux	11	Noticeable dimming
Partial (99% covered)	1,000 lux	8	Overcast-equivalent
Diamond ring / Baily's beads	10–100 lux	4–6	Remove solar filter
Totality (corona)	0.5–5 lux	0–3	Wide exposure bracket needed

Table 1. Typical illuminance and EV values at each solar eclipse stage.

The rate of illuminance change near C2 and C3 is highly non-linear. During most of the partial phase (C1 to roughly 10 minutes before C2), the dimming is gradual and almost imperceptible—from 100,000 lux to perhaps 20,000 lux over 50 minutes. But in the final 5 minutes before C2, illuminance plummets from ~5,000 lux to ~1 lux: a factor of 5,000× in 300 seconds. This means the exposure must react on a cadence of approximately 5 seconds during the ramp period, adjusting by 1–2 EV stops per update.

The practical consequence is a "twilight zone" of approximately 30 seconds on either side of C2/C3 where manual exposure management is simply impractical. The illuminance is changing too rapidly for a human to meter, evaluate, and adjust before the scene has shifted by another 2–3 stops. This is the window where automated exposure ramping provides its greatest value.

> **Non-linear Rate of Change**
>
> The illuminance curve near C2/C3 is approximately exponential, not linear. Fixed-interval exposure steps (e.g., "adjust 1 stop every 30 seconds") are insufficient because the required adjustment rate *accelerates* as totality approaches. EclipsePhoto must implement an accelerating ramp that increases the step frequency from once per minute (early partial) to once per 3–5 seconds (final approach to C2).

4 Eclipse Stage Detail: What Happens Physically

Each eclipse contact point involves specific physical phenomena that produce distinct photographic opportunities. Understanding what is visible at each stage—and for how long—drives both the exposure plan and the frame rate required to capture transient events.

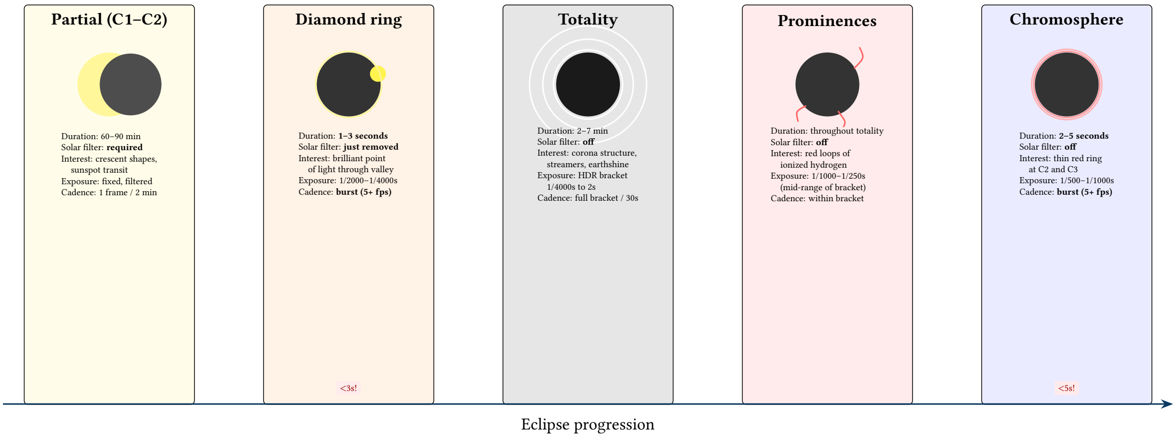


Figure 7. Eclipse stage detail: each phase presents distinct photographic targets with dramatically different durations and exposure requirements. Diamond ring and chromosphere events last only seconds and require burst-mode capture with pre-configured settings.

> **Transient Events**
>
> The diamond ring (1–3s) and chromosphere flash (2–5s) are the most time-critical moments of the eclipse. Missing them means waiting for the next eclipse. EclipsePhoto should pre-arm burst capture at these transition points with exposure settings locked to the expected brightness.

> **: Transient Event Pre-Arming**
>
> Diamond ring and chromosphere burst capture must be pre-armed before C2—the system cannot react fast enough to detect and configure for a 1–3 second event in real time. EclipsePhoto pre-loads burst parameters (shutter 1/2000–1/4000s, continuous drive) and triggers on the GPS-derived C2 timestamp.

4.1 Baily's Beads

In the seconds immediately before and after totality, the Moon's irregular limb profile—shaped by mountains, craters, and valleys—breaks the remaining sliver of photosphere into a chain of brilliant points known as Baily's beads. As the Moon advances, individual beads wink on and off as the photosphere is alternately blocked and revealed by lunar topography. The final, brightest bead becomes the diamond ring.

Baily's beads are photographically distinct from the diamond ring: where the diamond ring is a single dominant point with a coronal ring, beads appear as a necklace of 5–15 points distributed along the lunar limb. They are visible for approximately 5–15 seconds at C2 and again at C3, making them longer-lived than the diamond ring but still firmly in the "transient event" category.

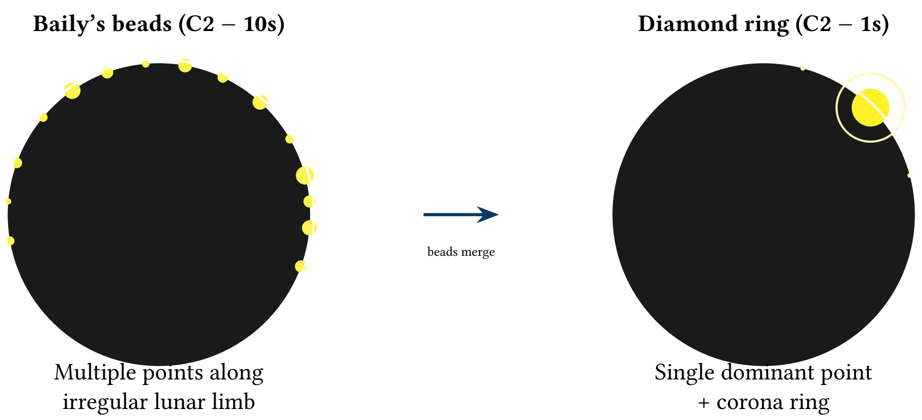


Figure 8. Baily's beads to diamond ring transition: as the Moon's limb covers the last arc of photosphere, sunlight shining through lunar valleys creates a chain of brilliant points (left). These merge into a single dominant bead—the diamond ring (right)—in the final 1–3 seconds before C2.

Target	Shutter	ISO	Duration	Notes
Baily's beads	1/2000s	100	5–15 s	Burst mode; same exposure as diamond ring
Diamond ring	1/4000s	100	1–3 s	Single dominant bead; risk of blooming

Table 2. Baily's beads and diamond ring exposure parameters. Both require burst-mode capture with fast shutter speeds to freeze the brilliant points without sensor blooming.

: Baily's Beads Capture Window

Baily's beads appear *before* the diamond ring at C2 and *after* it at C3. The burst capture sequence should begin 10–15 seconds before C2 (not at C2 itself) to catch the full bead sequence. At C3, continue burst capture for 10–15 seconds after the diamond ring fades.

4.2 The Diamond Ring

The diamond ring effect occurs in the final 1–3 seconds before C2 and again in the first 1–3 seconds after C3. It is created when the very last (or very first) sliver of photosphere shines through a single deep lunar valley while the rest of the solar disc is completely covered. The result is a single, intensely brilliant point of light on the lunar limb, surrounded by the faintly glowing corona—visually resembling a diamond set in a ring.

The diamond ring is vastly brighter than the surrounding corona—roughly 1,000–10,000×—which makes it both spectacular and technically challenging. The "diamond" itself is unfiltered photosphere (approximately 100,000 lux per unit area), and even though it occupies a tiny fraction of the solar limb, it can easily bloom across sensor pixels at slow shutter speeds. A fast shutter (1/2000–1/4000s) freezes the diamond cleanly; a slower shutter produces a dramatic starburst but risks clipping and blooming that obliterates the coronal ring.

At C2, the diamond ring signals the moment to remove the solar filter—the corona is only visible once the diamond fades. At C3, its reappearance signals filter replacement. These transitions are the most operationally critical moments of the entire eclipse (Section 9).

Two Diamond Rings

Every total eclipse produces *two* diamond rings: one at C2 (last light through the limb) and one at C3 (first light returning). The C3 diamond ring is often more dramatic photographically, because the operator has had the entire totality to settle equipment and refine exposure, and because the eye is dark-adapted. Both rings are unique—they occur at different points on the lunar limb where different valleys happen to be positioned.

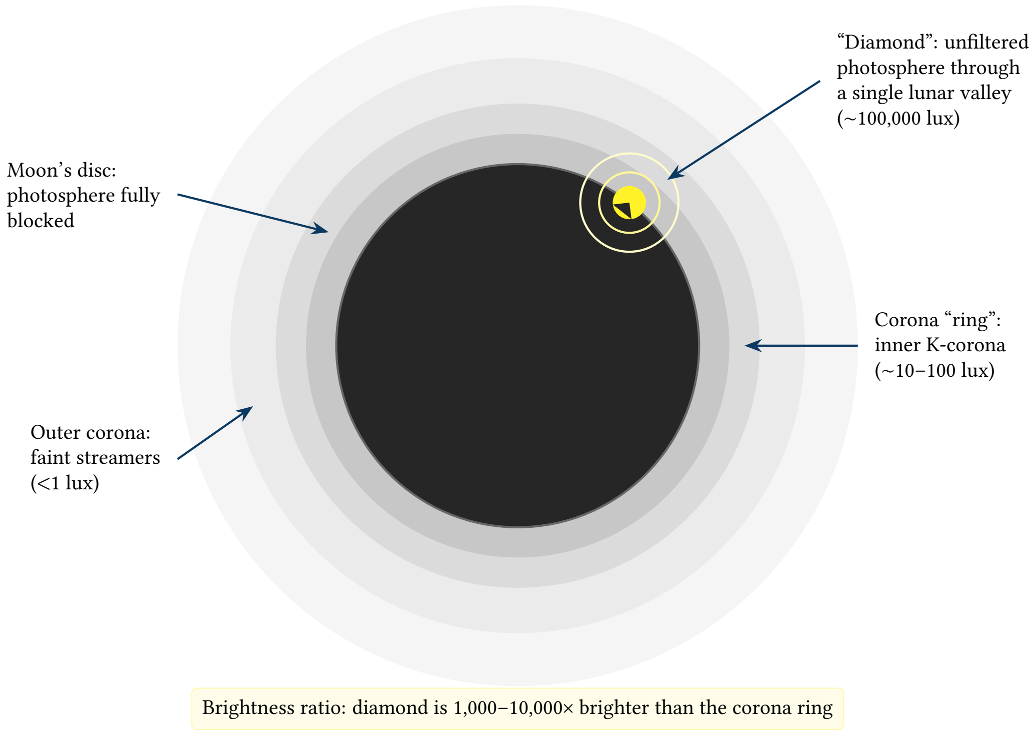


Figure 9. Diamond ring anatomy: the "diamond" is a point of unfiltered photosphere shining through a deep lunar valley, surrounded by the faint corona "ring." The extreme brightness contrast (>1,000×) makes this both visually spectacular and technically challenging—a fast shutter is essential to prevent sensor blooming.

4.3 Prominences

Solar prominences are vast arcs and loops of ionised hydrogen gas suspended above the Sun's surface by magnetic field lines. They glow at a characteristic deep red wavelength (Hα, 656.3 nm) and extend from the solar limb to heights of 10,000–150,000 km—roughly 1–20% of the solar diameter. During totality, prominences are visible as vivid red-pink features against the black lunar disc and the pearly white corona.

Prominences vary enormously from eclipse to eclipse. During periods of high solar activity (near solar maximum), large eruptive prominences may arc dramatically above the limb, creating some of the most striking images in all of astronomy. During solar minimum, prominences tend to be smaller and fewer, though they are rarely absent entirely. Checking solar activity data in the days before the eclipse (via NASA's Solar Dynamics Observatory imagery or NOAA's SWPC) gives a preview of what prominences will be visible and where on the limb they are positioned.

Photographically, prominences sit in the middle of the HDR bracket range. They are much fainter than the diamond ring or inner corona but much brighter than the outer corona. Typical exposure settings of 1/500–1/1000s at ISO 100–200 on a 400–600 mm lens capture prominence detail well. Because prominences are visible throughout totality (they do not change significantly over minutes), there is no urgency—they can be captured as part of each bracket cycle.

: Prominence Framing

Do not frame the corona rig too tightly on the solar disc. Prominences extend above the limb, and during active solar periods a large eruptive prominence can reach 0.5 solar radii (~5 arcminutes) above the limb edge. Leave at least 1.5 solar radii of clearance from the disc centre to the frame edge in all directions (see Section ??).

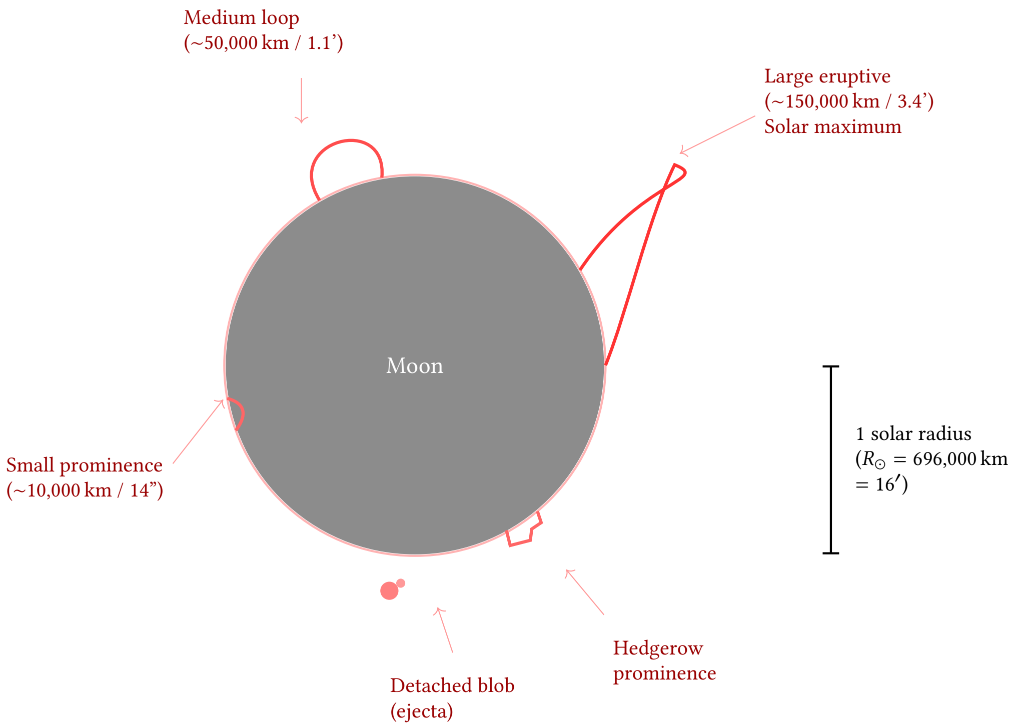


Figure 10. Prominence types and scale: prominences range from small quiescent loops (10,000 km, barely visible at 200 mm) to large eruptive arcs (150,000 km, spanning several arcminutes above the limb). Hedgerow prominences and detached blobs add further variety. The prominence population varies strongly with the solar cycle—near solar maximum, large eruptive prominences are common and demand generous framing.

4.4 Chromosphere Flash

The chromosphere is a thin layer of the solar atmosphere lying just above the photosphere (the visible "surface"). It is normally invisible—overwhelmed by the photosphere's brilliance—but during the 2–5 seconds immediately after C2 and before C3, when the photosphere is fully blocked but the chromosphere is not yet covered (or has just been uncovered), it briefly appears as a vivid red-pink ring encircling the entire lunar limb. This is the chromosphere flash.

The chromosphere's red colour comes from the same Hα emission (656.3 nm) that illuminates prominences—the chromosphere and prominences are physically the same plasma layer, but the chromosphere flash shows the emission from the full limb circumference rather than from isolated loops. The flash is thin (only a few arcseconds wide) and extremely brief, making it one of the most technically demanding eclipse targets.

Capturing the chromosphere flash requires:

- **Precise timing:** The flash lasts 2–5 seconds at most. EclipsePhoto must begin burst capture within 1 second of C2 to catch it.
- **Fast shutter:** 1/500–1/1000s freezes the thin ring without motion blur.
- **Sufficient focal length:** At 200 mm the chromosphere ring is only ~1 pixel wide; 400 mm+ is needed to resolve it as a distinct feature.
- **Burst mode:** Continuous shooting at 5+ fps maximises the chance of capturing the flash at its peak intensity before it fades as the Moon fully covers it.

The chromosphere flash at C3 is the reverse process: as the Moon begins to uncover the chromosphere, the red ring reappears briefly before being overwhelmed by the returning photosphere. The C3 flash is typically captured as part of the same burst sequence that captures the C3 diamond ring.

> **Chromosphere vs. Prominences**
>
> Prominences and the chromosphere flash are physically related—both are Hα emission from ionised hydrogen—but they differ in visibility window and capture strategy. Prominences are visible throughout totality and can be captured at leisure within each bracket cycle. The chromosphere flash exists only in the 2–5 second windows at C2 and C3 and requires dedicated burst capture. EclipsePhoto treats them as separate targets with separate exposure profiles.

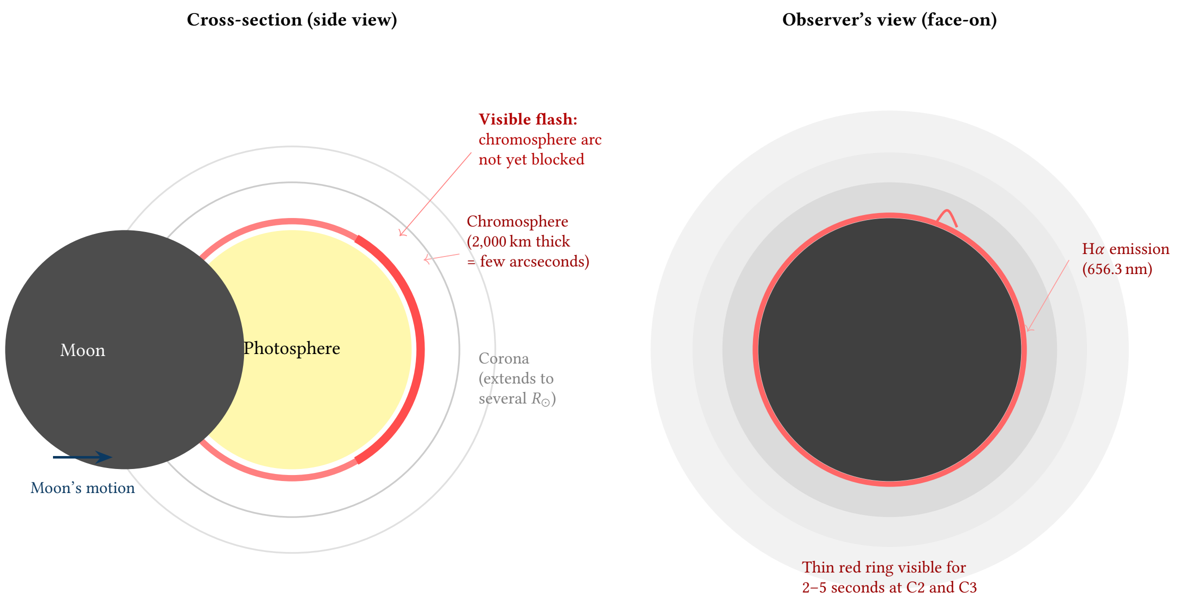


Figure 11. Chromosphere flash: the cross-section (left) shows the thin chromosphere layer sandwiched between the photosphere and corona. As the Moon blocks the photosphere but not yet the chromosphere, a thin red arc is briefly visible. The observer's view (right) shows the resulting vivid red ring encircling the lunar limb—visible for only 2–5 seconds at C2 and C3.

4.5 Near-Totality Atmospheric Phenomena

Several atmospheric and environmental phenomena occur in the minutes surrounding totality. While secondary to the solar disc itself, they provide compelling photographic subjects—especially for the wide-angle rig—and serve as visual cues for the operator.

Shadow Bands. In the 1–2 minutes immediately before C2 and after C3, faint rippling bands of light and shadow may appear on light-coloured surfaces (white sheets, concrete, car roofs). These are caused by atmospheric turbulence refracting the thin remaining crescent of sunlight, analogous to the shimmering patterns at the bottom of a swimming pool. Shadow bands are subtle, low-contrast, and fast-moving (~1–3 m/s), making them difficult to photograph. A wide-angle lens pointed at a light surface, shooting video at high frame rate (60–120 fps), provides the best chance. Not every eclipse produces visible shadow bands—they depend on atmospheric conditions.

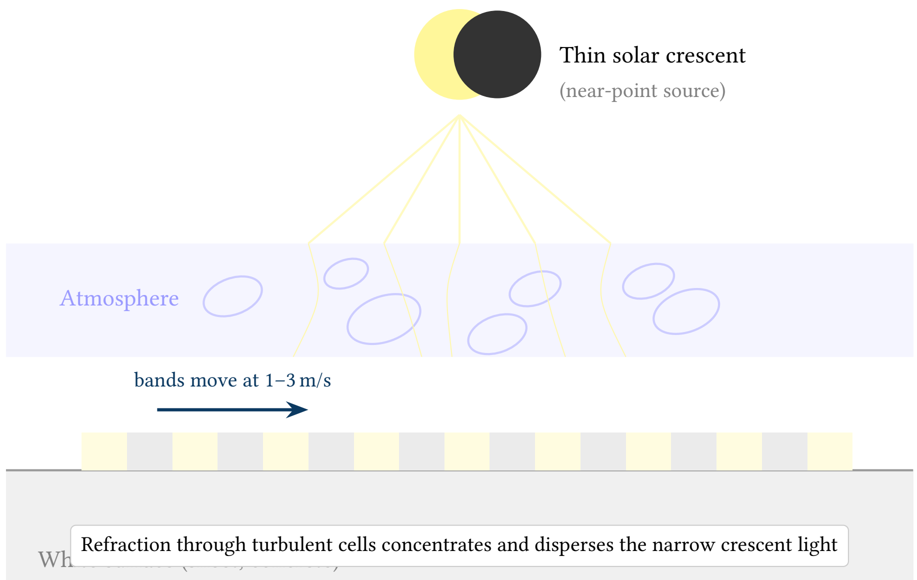


Figure 12. Shadow band formation: the thin solar crescent near C2/C3 acts as a near-point source. Atmospheric turbulence cells refract the light unevenly, creating rapidly moving bands of light and shadow on flat surfaces below—similar to the ripple patterns at the bottom of a swimming pool.

Ground-Level Scintillation and Shimmering. In the minutes surrounding totality, the landscape itself appears to shimmer—an effect superficially similar to heat haze above a sun-baked road, but driven by a fundamentally different mechanism. As the umbral shadow races across the Earth's surface at 1,500+ km/h, its sharp illumination boundary creates extreme temperature differentials in the lower atmosphere: the ground cools rapidly under the shadow while adjacent sunlit areas only metres away remain warm. This establishes turbulent convection cells and density fluctuations in the air column on spatial scales of centimetres to metres. Because the refractive index of air depends on both temperature and density ($n - 1 \propto \rho/T$), the turbulent mixing introduces optical path length differences across adjacent sightlines, manifesting as visible shimmering and distortion of distant objects.

The effect is more dramatic than ordinary heat shimmer because the temperature gradient is imposed so rapidly—ambient temperature drops 3–8°C in minutes—and because the illumination boundary is geometrically sharp rather than diffuse. The scintillation is most noticeable when looking across flat terrain or water toward the approaching or receding shadow edge, where the observer's line of sight traverses the maximum extent of the disturbed air column. Video recording of distant objects (trees, buildings, horizon features) along the shadow boundary at 60+ fps captures the effect effectively; still photography rarely conveys the dynamic character.

The rapidly changing light also triggers biological responses across the local environment: birds return to roost, crickets and cicadas begin their evening chorus, flowers close their petals, and domesticated animals exhibit confusion. While these responses are driven by the light change rather than the atmospheric scintillation per se, they contribute to the uncanny ground-level sensory experience that distinguishes totality from a simple darkening of the sky.

The Approaching Umbral Shadow. In the 30–60 seconds before C2, observers looking toward the west-northwest (in the Northern Hemisphere) can see the Moon's shadow racing across the landscape at 1,500–3,000 km/h, appearing as a dark wall sweeping toward them. After C3, the shadow recedes to the east-southeast. This is best captured with a wide-angle lens or video camera pointed at the distant horizon, not at the Sun.

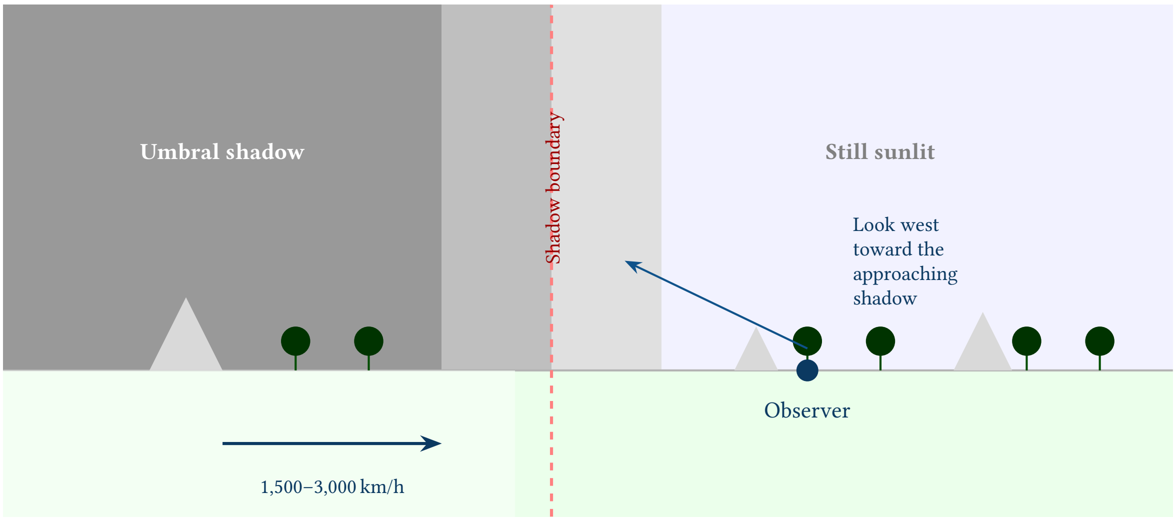


Figure 13. The approaching umbral shadow: 30–60 seconds before C2, the Moon's shadow appears as a dark wall racing across the landscape from the west at 1,500–3,000 km/h. The boundary between shadow and sunlight is visible on the ground and sky. Best captured with a wide-angle lens pointed at the horizon, not the Sun.

360° Horizon Glow. During totality, the entire horizon exhibits a 360° sunrise/sunset glow. This occurs because observers in the umbral shadow are surrounded by regions 100–200 km away that are still in partial eclipse and receiving sunlight. The glow is typically strongest in the direction opposite the shadow's approach. The wide-angle rig captures this naturally as part of the landscape context.

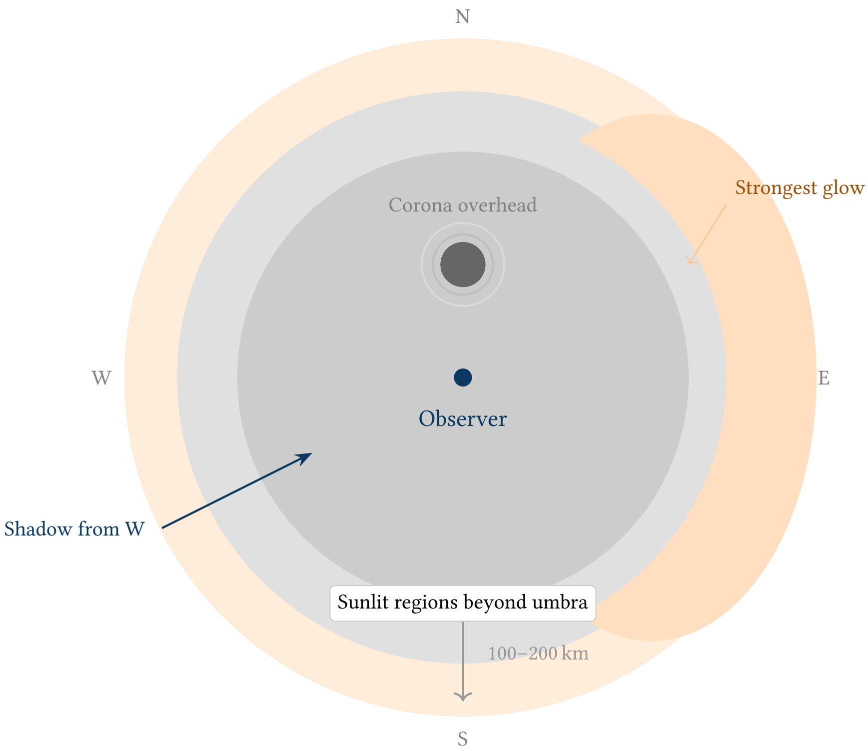


Figure 14. 360° horizon glow during totality: the observer stands in the umbral shadow while regions 100–200 km away in every direction are still sunlit, producing a continuous sunset-like glow around the entire horizon. The glow is strongest opposite the shadow's approach direction (here, to the east).

Planets and Bright Stars. When totality darkens the sky to roughly equivalent to deep twilight (~1 lux), planets and bright stars become visible. Venus is almost always visible during totality; Jupiter, Mars, and Mercury may also appear depending on their positions. Bright stars like Sirius or Regulus can emerge. These add context to wide-angle images and can serve as focus confirmation targets if pre-identified.

Temperature Drop and Wind. Ambient temperature typically drops 3–8°C during the final partial phases, with the minimum occurring 1–2 minutes after C2 (thermal lag). An associated drop in wind speed or shift in wind direction often accompanies the temperature change. These are not photographic subjects per se, but affect equipment (see focus drift in Section 15) and operator comfort.

Figure 15. Near-totality phenomena timeline: six categories of observable events mapped around C2 and C3. Shadow bands, Baily's beads, and the approaching shadow occur in the minutes before and after totality; corona, prominences, horizon glow, and environmental effects persist throughout.

4.6 Star and Planet Visibility During Eclipse

During totality, sky brightness drops to approximately 1–10 lux—equivalent to deep civil twilight—corresponding to a sky surface brightness of roughly magnitude 3–4 per square arcsecond. Under these conditions, celestial objects brighter than approximately magnitude +2 to +3 become visible to the naked eye.

Visibility Threshold Curve. The emergence of stars and planets follows a predictable curve tied to the declining illumination in the final minutes before C2. Venus (apparent magnitude −4.0 to −3.9) is the first object to appear, typically becoming visible 10–15 minutes before C2 while the partial phase is still well advanced. Jupiter (magnitude −2 to −1.8) may emerge 2–5 minutes before C2 as the sky darkens more rapidly. First-magnitude stars—Sirius (−1.46), Canopus (−0.74), Arcturus (−0.05)—become visible within seconds of C2 as the photosphere is fully extinguished. During deep totality, stars as faint as magnitude +2 to +3 may be glimpsed if totality duration exceeds ~2 minutes and the atmosphere is transparent.

The reverse process occurs after C3: objects disappear in reverse order of brightness as the sky brightens, with fainter stars vanishing first and Venus persisting for several minutes into the partial phase.

Object	Apparent Mag	Approx. Visibility Threshold	Expected Visibility Window
Venus	−4.0	Pre-C2 by 10–15 min	Throughout totality + partial phases
Jupiter	−1.8	Pre-C2 by 2–5 min	Most of totality
Sirius	−1.46	~C2	Totality only
Canopus	−0.74	~C2	Totality only
Arcturus	−0.05	~C2	Totality only
Regulus	+1.4	~C2 + 10 s	Deep totality only
Castor	+1.6	~C2 + 15 s	Deep totality only
Bellatrix	+1.6	~C2 + 15 s	Deep totality only

Table 3. Predicted naked-eye visibility of stars and planets during totality. Actual thresholds depend on sky transparency, corona brightness, angular distance from the Sun, and totality duration. Objects within ~15° of the Sun may be washed out by coronal glow regardless of their intrinsic brightness.

Factors Affecting Visibility. Which stars and planets are actually visible during any given eclipse depends on several factors:

- **Angular distance from the Sun.** Objects within approximately 15° of the Sun are washed out by the coronal glow, regardless of their intrinsic brightness. This is analogous to the difficulty of seeing stars near a bright streetlight.
- **Date and time of eclipse.** The Sun's position on the ecliptic determines which constellations and planets lie nearby. An eclipse in August places the Sun in Leo, making Regulus a likely candidate; an eclipse in December puts it in Sagittarius near the galactic centre.
- **Atmospheric transparency.** Haze, thin cloud, and aerosol loading raise the sky background brightness and reduce the number of visible objects. A high-altitude, dry site yields the deepest sky darkening.
- **Totality duration.** Longer totality produces a darker sky at mid-eclipse, pushing the visibility threshold to fainter magnitudes. A 4-minute totality may reveal stars a full magnitude fainter than a 1-minute totality.

What "Deep Totality" Means. The term *deep totality* refers to the period around mid-eclipse when the sky reaches its maximum darkness. Even after C2, the sky does not instantly drop to its darkest level—it continues to darken for 30–90 seconds as the umbral shadow fully envelops the observer's local atmosphere.

The reason is atmospheric scattering. The sky's brightness during totality is not determined solely by direct sunlight (which is fully blocked) but by light scattered from the upper atmosphere above regions *outside* the umbral shadow. At C2, the shadow has just arrived: sunlit atmosphere is still relatively close on all sides, only 50–80 km away, and scattered light from these regions keeps the sky brighter than it will eventually become. As the shadow moves overhead and the observer moves deeper into the centre of the umbral cone, the nearest sunlit atmosphere recedes to 100–200 km in every direction. The scattering contribution from these distant regions is weaker (it falls off roughly with the inverse square of distance and is further attenuated by the atmospheric path), so the sky continues to darken.

For a short totality of 1 minute, the shadow passes so quickly that the observer never reaches the deepest point—the sky begins brightening again almost as soon as it finishes darkening. For a long totality of 4–7 minutes, the observer spends 2–4 minutes near the shadow centre, where sky brightness can drop an additional 0.5–1.0 magnitude per square arcsecond below the C2 level. This is deep totality: the window when the faintest stars emerge and the outer corona is most visible against the darkest possible sky background.

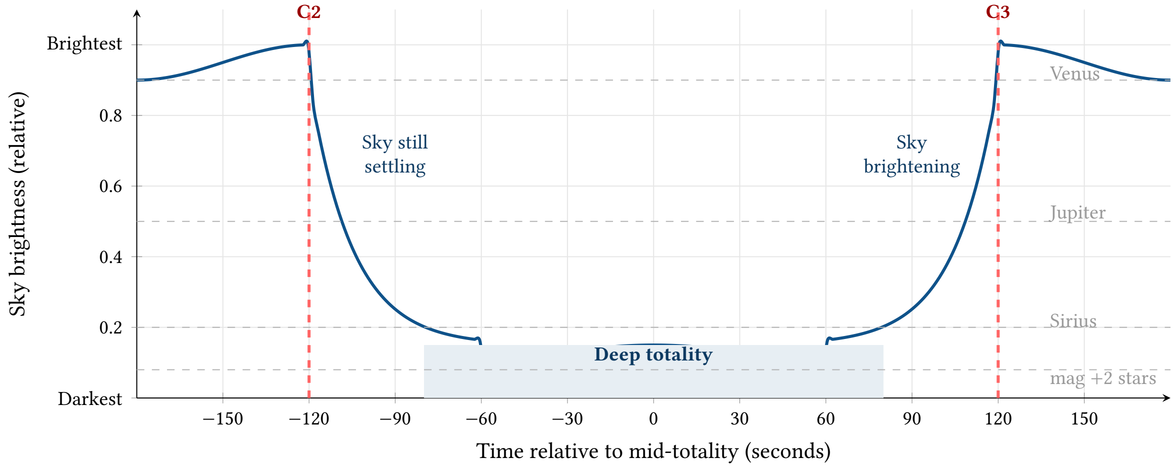


Figure 16. Sky brightness during a 4-minute totality (C2 at −120 s, C3 at +120 s). After C2, scattered light from the nearby sunlit atmosphere takes 30–60 seconds to fade as the umbral shadow moves overhead. The darkest sky—deep totality—occurs around mid-eclipse, when the nearest sunlit atmosphere is 100–200 km away in every direction. Horizontal dashed lines show the approximate sky brightness threshold at which different objects become visible. Faint stars (magnitude +2) are only visible during deep totality; Venus is visible well before C2.

The practical consequence for EclipsePhoto is that the faintest targets—outer corona streamers and earthshine on the lunar disc—are best captured during the deep totality window, when the sky background is minimised. The bracket sequence should prioritise long exposures (2–4 s for outer corona) during mid-totality, saving the burst-mode transient captures (diamond ring, chromosphere) for the C2 and C3 boundaries where they actually occur.

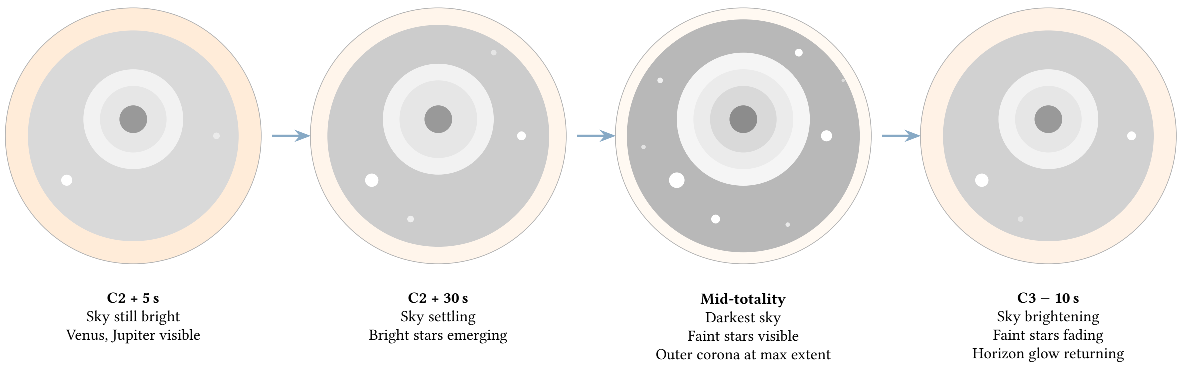


Figure 17. Sky appearance at four stages through a 4-minute totality. At C2 + 5 s (left) the sky is still relatively bright, with only Venus and Jupiter visible. By C2 + 30 s, bright stars are emerging as scattered light fades. At mid-totality (centre-right), the sky reaches maximum darkness: the faintest stars appear and the outer corona extends to its full visible radius. Approaching C3 (right), the sky begins brightening and faint objects disappear. This progression is why long totalities produce better corona images and reveal more stars.

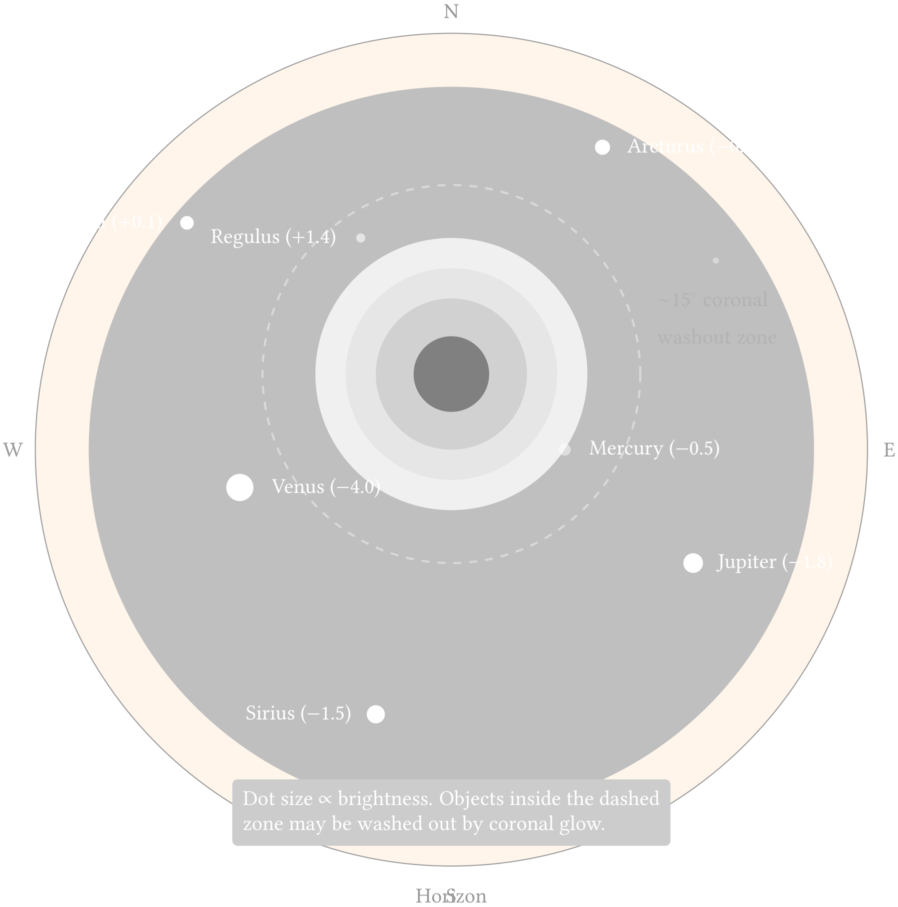


Figure 18. Simulated eclipse sky chart: during totality, the sky darkens to deep twilight, revealing planets and bright stars around the eclipsed Sun. Venus (magnitude −4.0) is almost always visible; Jupiter and bright stars appear at or just after C2. Objects within ~15° of the Sun (dashed circle) may be hidden by coronal glow. Star/planet positions are illustrative—actual positions depend on the eclipse date.

: Visible Object Prediction

EclipsePhoto should compute and display predicted visible objects based on the eclipse date, location, and current planetary ephemeris. Pre-computing a sky chart overlay for the wide-angle rig's field of view aids both planning (identifying focus-confirmation targets) and post-processing (identifying objects in captured frames).

4.7 Solar Altitude and Atmospheric Effects

The Sun's altitude above the horizon at the time of totality significantly affects corona appearance and exposure requirements. When the Sun is low in the sky, its light must travel through a thicker layer of atmosphere to reach the camera. This longer path dims the corona, shifts its colour toward red, and reduces the visibility of faint outer structures—the same effect that makes sunsets red and dim compared to midday sun.

How Much Atmosphere Is in the Way? The term *air mass* describes how much atmosphere the light passes through, relative to looking straight up. Looking directly overhead (zenith), the air mass is 1.0—the minimum possible. As the Sun drops toward the horizon, the light path tilts and cuts through progressively more atmosphere:

Solar Altitude	Zenith Angle	Air Mass X
90°	0°	1.0
60°	30°	1.15
45°	45°	1.41
30°	60°	2.0
20°	70°	2.9
15°	75°	3.8
10°	80°	5.6

Low-Altitude Eclipses (Sun Below 30°). When the Sun is low, it is viewed through more than twice the normal atmospheric thickness. The corona appears warmer and redder—the same atmospheric scattering that makes sunsets orange preferentially strips out blue light from the corona. The faint outer corona is dimmed disproportionately, reducing its visible extent and contrast. Slight colour fringing may also appear in high-magnification images. The overall effect is a "warm sunset" quality to the corona that, while visually appealing, complicates white balance calibration and HDR compositing.

High-Altitude Eclipses (Sun Above 60°). With air mass below 1.15, atmospheric effects are minimal. The corona appears in its natural cool white tones, contrast is maximised, and the outer corona extends to its full visible radius. These conditions are optimal for scientific-quality imaging and simplify post-processing because atmospheric colour shifts are negligible.

Exposure Implications. Each doubling of the atmospheric path roughly costs 0.3–0.5 EV in brightness. For a low-altitude eclipse at 15° elevation (air mass 3.8), the outer corona may require 1.0–1.5 EV longer exposure compared to a high-altitude baseline. The bright inner corona is less affected in practical terms, but the bracket sequence must extend further on the long-exposure end to capture the dimmed outer structures.

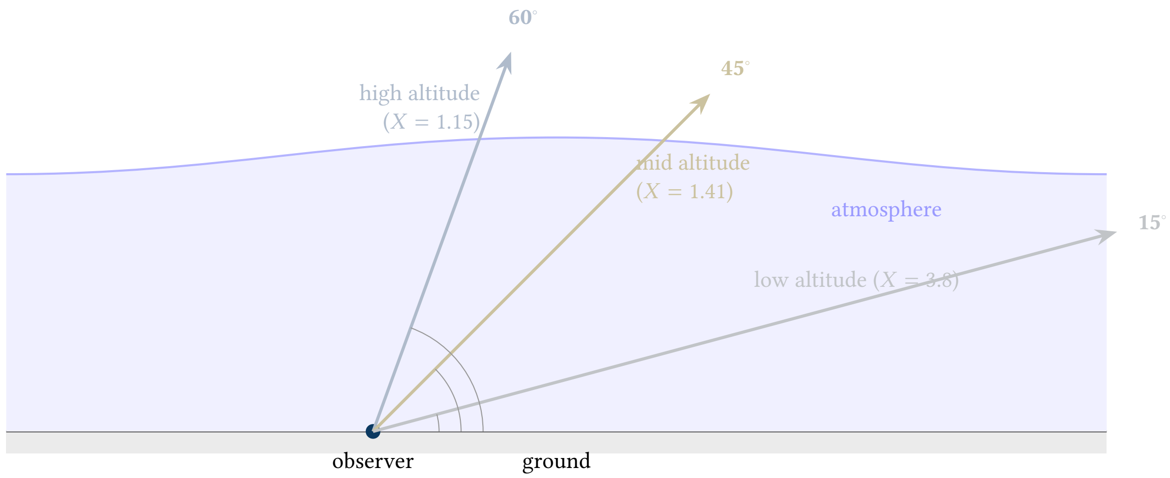


Figure 19. Atmospheric path length at different solar altitudes. At 15° elevation the light passes through 3.8× as much atmosphere as when the Sun is overhead, dimming and reddening the corona. High-altitude eclipses (>60°) minimise these effects.

Site Selection Implications. When choosing an observation site along the eclipse path, solar altitude should be factored alongside weather probability, accessibility, and totality duration. Eclipses occurring near sunrise or sunset (altitude < 15°) present dramatic horizon compositions but challenging photometric conditions. Sites where the eclipse occurs at maximum altitude along the path generally yield the highest-quality corona images.

: Solar Altitude Compensation

The system should compute solar altitude for the eclipse time and location, and adjust baseline exposure tables for atmospheric dimming. Display the solar altitude on the planning screen so the operator can factor it into site selection. For eclipses below 30° altitude, extend the HDR bracket sequence by 1–2 additional long-exposure frames to compensate for the dimmed outer corona.

5 Dynamic Range Challenge: Corona and Prominences

The solar corona alone spans approximately 12 EV stops of brightness—far exceeding the dynamic range of any single exposure. The inner corona (within 0.5 solar radii) is roughly 1 million times brighter than the outer corona (at 3+ solar radii). Capturing the full extent requires HDR bracketing with a wide spread of exposures.

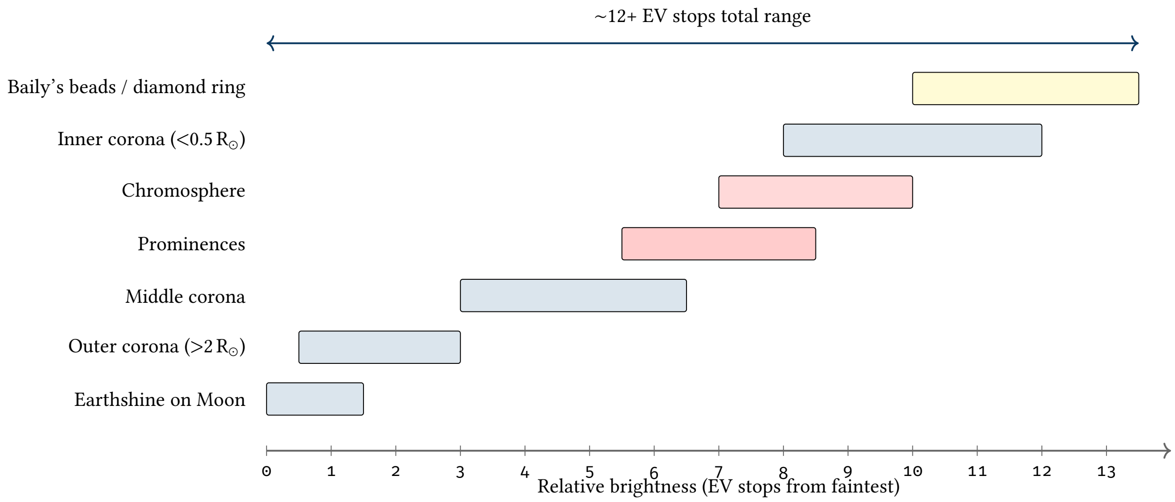


Figure 20. Dynamic range of eclipse features: from faint earthshine on the lunar surface to the brilliant diamond ring, the brightness span exceeds 12 EV stops.

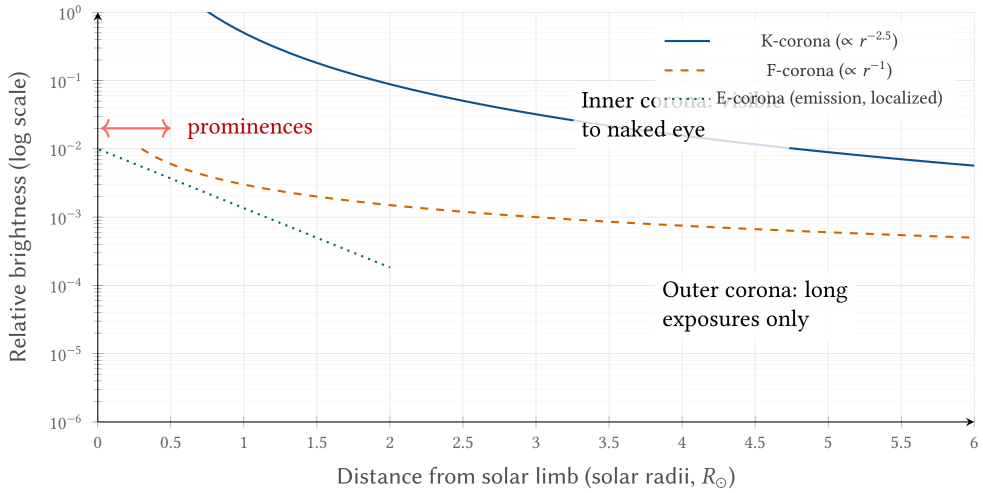


Figure 21. Corona radial brightness profile: the K-corona (electron-scattered) dominates the inner region and drops steeply ($\sim r^{-2.5}$), while the F-corona (dust-scattered) dominates at larger distances with a shallower decline. This radial gradient is the physical reason the dynamic range exceeds 12 EV stops.

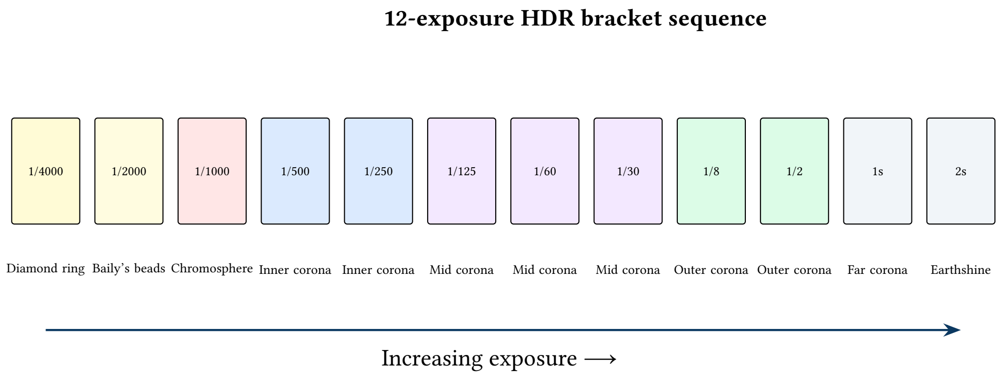


Figure 22. Bracketing strategy: a 12-exposure sequence from 1/4000s to 2s, with each exposure targeting a specific eclipse feature. The full bracket captures the complete dynamic range for HDR compositing.

5.1 Solar Activity and Prominence Monitoring

Insight: *The corona alone spans ~12 EV stops—no single exposure captures it. Automated HDR bracketing at totality onset is the single most critical automated action.*

Before the eclipse, the operator should consult solar activity data from NOAA's Space Weather Prediction Center (SWPC) and NASA's Solar Dynamics Observatory (SDO) to anticipate prominence activity. Active regions near the solar limb may produce prominences visible during totality—and their size can vary dramatically, from small loops barely extending beyond the limb to exceptional eruptions reaching 0.5 solar radii or more.

Prominences are plasma structures suspended in the corona by magnetic fields. Their height above the solar limb determines whether they will be visible to the naked eye during totality (most are) and whether they will fit within the frame of a tightly cropped telephoto image (some will not).

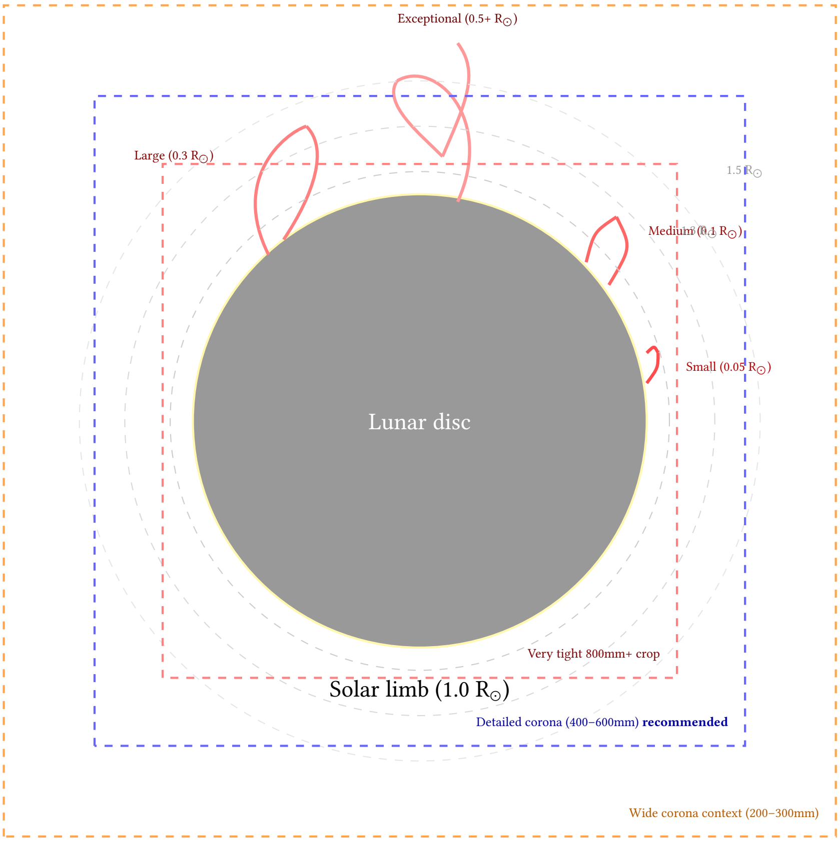


Figure 23. Prominence heights relative to the solar disc: small prominences (0.05 $R_\odot$) are common and fit any framing; exceptional eruptions (0.5+ $R_\odot$) can extend well beyond a tight telephoto crop. The red dashed box shows the risk of 800 mm+ framing; the blue dashed box shows the recommended 400–600 mm range for detailed corona work; the orange dashed box shows the wider 200–300 mm context framing.

Framing for Prominences. The primary corona rig should use a focal length in the 400–600 mm range on a full-frame sensor. This is the productive sweet spot for detailed corona photography: the solar disc is large enough to reveal prominence structure and inner coronal detail, while the outer corona (to ~2–3 solar radii) fits comfortably within the frame. At the author's first eclipse (2024), a 600 mm full-frame setup captured the majority of the corona successfully, confirming this range as practical.

At 200–300 mm on full-frame, the entire corona with extended streamers fits easily, but the disc is small—roughly 10–15% of frame width. This is the appropriate range for a *context rig* providing a generous wide-corona view, but it sacrifices the detail that makes telephoto eclipse photography compelling.

At 600–800 mm, corona photography works well with careful framing, but the risk of cropping outer streamers and large prominences increases. Operators should consult pre-eclipse solar activity data (see above) to assess whether active regions near the limb are likely to produce large prominences. If significant limb activity is expected, err toward the shorter end of this range.

Beyond 800 mm, framing becomes very tight and significant portions of the corona are cropped. These focal lengths are best reserved for a dedicated close-up or prominence rig—not as the primary corona rig. Solar maximum periods (such as 2024–2026) produce more frequent and larger prominences, making this consideration especially important.

The recommended approach is to frame the primary corona rig at a focal length that provides at least 2.5 $R_\odot$ of clearance beyond the limb in all directions. At 500 mm on a full-frame sensor, this is comfortably achieved while still resolving fine prominence and coronal structure.

5.2 Framing Assistant Tool

EclipsePhoto should include a framing calculator and simulator that shows operators exactly what the camera will see at a given focal length and sensor size combination. Pre-eclipse planning with this tool eliminates guesswork about lens selection and rig assignment.

Inputs. The framing assistant accepts focal length (mm) and sensor dimensions. Presets are provided for common sensor formats:

- **Full-frame:** 36×24 mm (e.g. Sony A7-series, Canon R-series, Nikon Z-series)
- **APS-C:** 23.5×15.6 mm (e.g. Sony A6000-series, Fujifilm X-series)
- **Micro Four Thirds:** 17.3×13 mm (e.g. OM System, Panasonic Lumix G-series)

Outputs. For each combination, the tool computes and displays:

- Field of view (horizontal, vertical, diagonal) in degrees
- Solar disc size as a percentage of frame width and height
- Corona extent coverage: how many solar radii from disc center fit within the frame
- Prominence clearance margin: the distance from the limb to the frame edge in $R_\odot$

Visual Overlay. The simulator renders a mockup frame with overlaid circles showing the solar disc, inner corona (1.5 $R_\odot$), outer corona (3 $R_\odot$), and a typical prominence scale marker. This gives an immediate visual sense of composition and cropping risk.

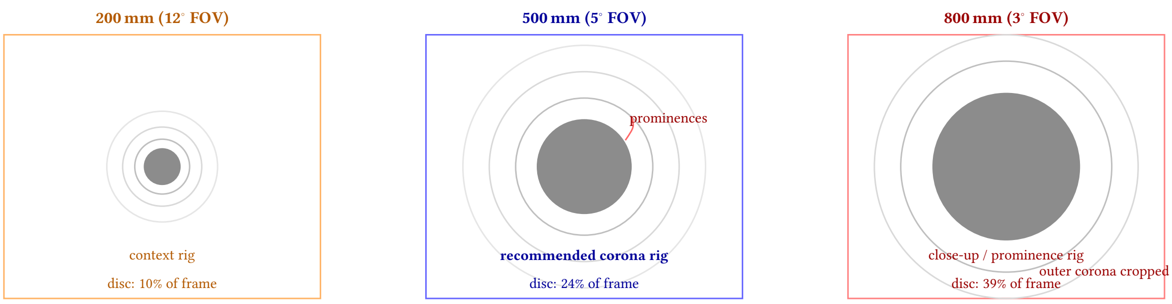


Figure 24. Framing assistant preview: three full-frame sensor views at different focal lengths. At 200 mm (left), the disc is small and ample sky surrounds the corona—suitable for a context rig. At 500 mm (center), the disc is well-sized for prominence detail while the outer corona fits comfortably—the recommended primary corona rig. At 800 mm (right), the disc fills the frame and outer corona is cropped—best reserved for a dedicated close-up rig.

: Pre-Eclipse Framing Check

Before each eclipse, run the framing simulator with the planned lens/camera combinations. Verify that the primary corona rig captures at least 2.5 solar radii from disc center in all directions. Check recent solar activity data to assess prominence risk and adjust framing if large limb prominences are expected.

5.3 Optical Filters for Eclipse Photography

Beyond the mandatory solar ND filter used during partial phases, photographers frequently ask about additional optical filters for eclipse work. This subsection examines three categories: narrowband hydrogen-alpha filters, broadband light pollution filters, and UV/IR cut filters.

5.3.1 Hydrogen-Alpha (Hα) Narrowband Filters

Hydrogen-alpha (Hα) narrowband filters isolate the 656.3 nm hydrogen emission line and can dramatically enhance prominence and chromosphere visibility. During totality, an Hα filter on a dedicated rig reveals fine prominence structure—filamentary arcs, detached blobs, and hedgerow prominences—that may be subtle or invisible in broadband white-light images. The chromosphere flash at C2/C3, a thin red ring visible for 2–5 seconds, is also significantly enhanced.

However, Hα filters carry substantial operational constraints:

- **Cost:** Dedicated solar Hα systems (Daystar Quark, Lunt, DayStar Quantum) typically cost \$1,000–\$4,000 for the filter or etalon alone, plus a compatible optical train.
- **Throughput:** The extremely narrow bandpass (~0.5–1.0 Å) transmits a tiny fraction of incident light, requiring much longer exposures than broadband capture. During totality, where photons are already scarce, this compounds the exposure challenge.

- **Optical train:** Hα etalon filters require specific focal ratios (typically f/30 for Daystar Quark systems), telecentric optics, or purpose-built refractors. They cannot simply be threaded onto a standard camera lens.
- **Dedicated rig requirement:** Given the above constraints, Hα capture effectively requires its own camera and optical system, separate from the broadband corona rig.
- **Diminishing returns during totality:** Prominences are readily visible in broadband light during totality—Hα provides enhancement and finer structural detail, not a capability that is otherwise absent.
- **Chromosphere flash window:** The 2–5 second chromosphere flash at C2/C3 can produce stunning Hα images, but the window is extremely brief and demands precise timing.

5.3.2 Light Pollution and Broadband Filters

Light pollution filters (e.g., Optolong L-Pro, IDAS LPS, Astronomik CLS) are designed for deep-sky astrophotography under urban skies, selectively blocking sodium and mercury emission lines from artificial lighting. They are *not* useful for eclipse photography:

- During partial phases, the sky is sunlit—light pollution filters have no meaningful effect on a daytime sky dominated by broadband solar illumination.
- During totality, the sky darkens to deep twilight levels. The corona itself is broadband white light (scattered photospheric continuum); a light pollution filter would attenuate the corona signal without removing any unwanted emission, reducing signal-to-noise ratio for no benefit.
- For remote or desert eclipse sites, artificial light pollution is negligible to begin with.
- For urban eclipse sites, the 2–7 minutes of totality darkening overwhelms any steady-state urban sky glow. The dominant "noise" source is scattered coronal and atmospheric light, not streetlamps.
- **Summary:** light pollution filters should be omitted entirely from eclipse photography kits.

5.3.3 UV/IR Cut Filters

Ultraviolet and infrared cut filters block wavelengths outside the visible band, preventing IR contamination that can reduce contrast and introduce a reddish color cast in some optical designs. Most modern Sony Alpha sensors include a built-in IR-blocking filter over the sensor stack, making an external UV/IR cut filter redundant. Older lens designs with lower-quality coatings may benefit from a UV/IR cut, but for contemporary camera–lens combinations, the effect is negligible. Unless using a modified (IR-removed) camera body, UV/IR cut filters are generally unnecessary.

> **: Filter Strategy During Totality**
>
> For the primary corona and close-up rigs, shoot broadband with no filter during totality. The corona's full spectral content and maximum photon throughput are essential for HDR bracket sequences spanning 12+ EV stops. Hα is a specialist addition for a dedicated prominence rig, not a replacement for broadband capture.

6 Color Temperature Progression During Eclipse

The correlated color temperature (CCT) of ambient light changes significantly during a solar eclipse. This affects camera auto white balance behavior and the appearance of both the eclipse itself and the surrounding landscape. EclipsePhoto should lock white balance rather than allowing auto adjustment.

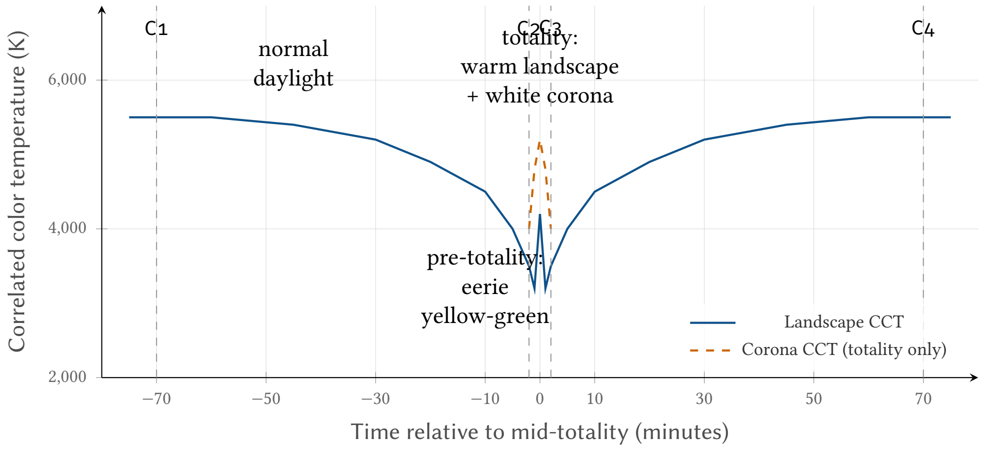


Figure 25. Color temperature progression: landscape CCT drops from ~5500 K (daylight) to ~3200 K just before totality as atmospheric scattering shifts toward warm tones. The corona itself is ~5000–5200 K (white). Auto white balance will oscillate—lock WB before the eclipse begins.

: White Balance Lock

Lock camera white balance to a fixed value (5200–5500 K daylight) before C1. Auto white balance will hunt during the rapid color temperature changes and produce inconsistent frames that complicate timelapse assembly. When shooting RAW (recommended), white balance is metadata-only and can be adjusted losslessly in post-processing—but locking it ensures consistent LCD preview and histogram readings during the event, and simplifies batch processing of timelapse sequences.

7 Focal Length and Field of View

Target selection drives focal length choice. The solar corona extends ~3° from the Sun's center, and deep-sky objects vary widely. A field-of-view diagram helps match lenses to targets and informs the 500-rule calculations in Section 8.

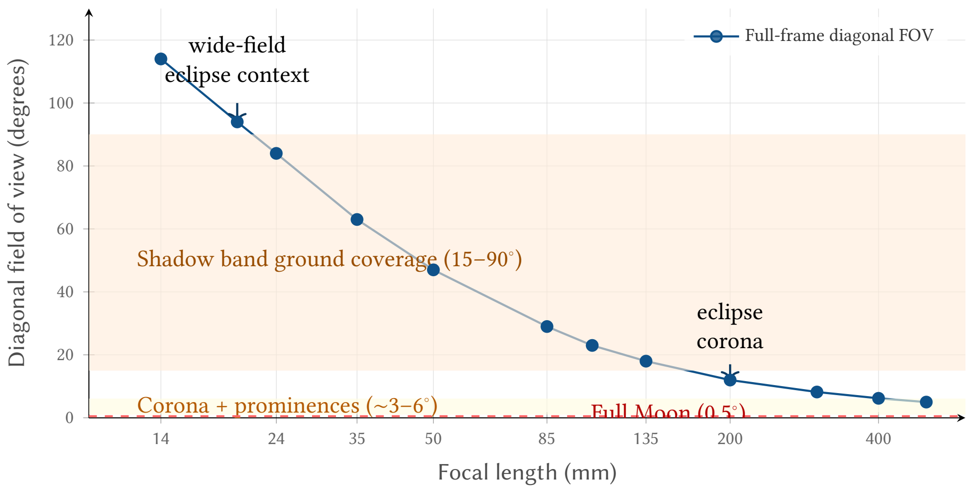


Figure 26. Focal length vs. diagonal field of view (full-frame sensor) with target angular sizes overlaid. Wide-angle lenses (14–35 mm) suit landscape context and shadow band capture; telephoto (400–600 mm) is the recommended range for detailed corona work and lunar surface.

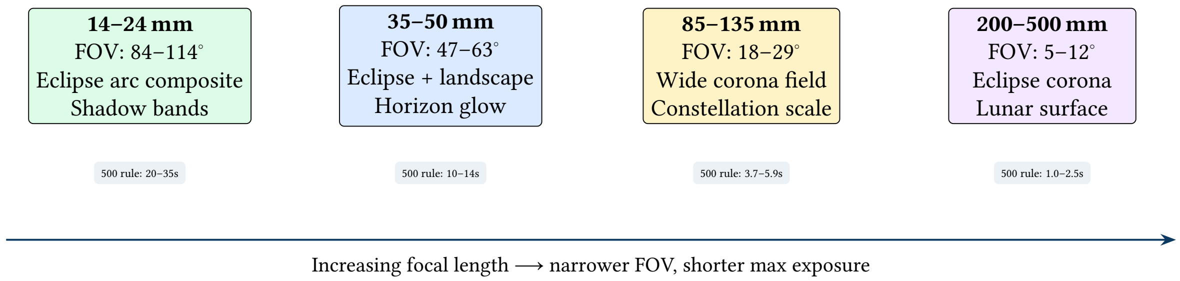


Figure 27. Lens focal length categories with target assignments and maximum untracked exposure times (500 rule). Longer focal lengths demand faster shutter speeds, compressing the available exposure range.

8 Camera Parameter Adaptation Strategy

The extreme dynamic range of eclipse events requires a systematic approach to camera parameter changes. The strategy prioritizes shutter speed adjustment (fastest response, preserves optical quality) over ISO changes (affects noise floor), while holding aperture fixed (maintains optical performance and depth of field).

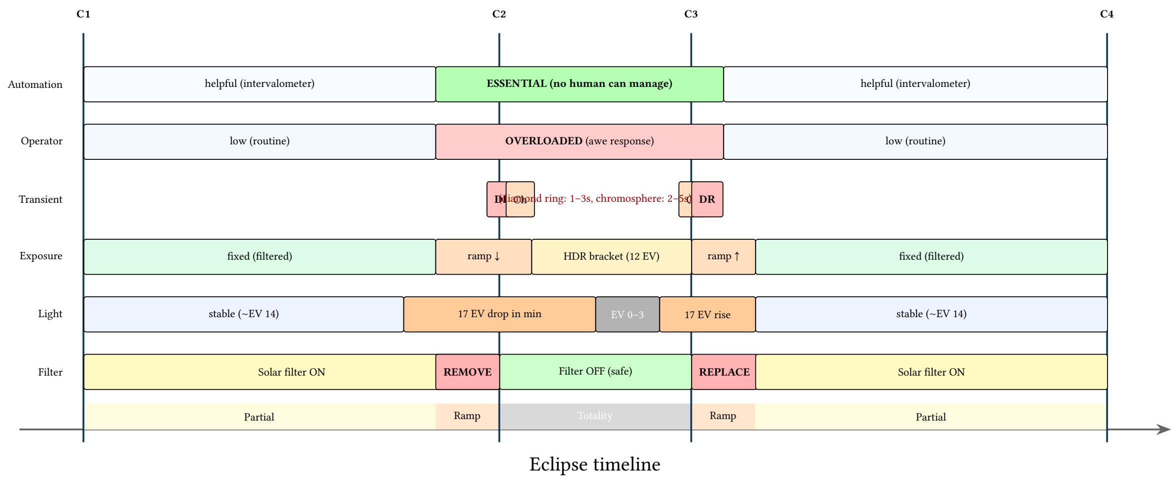


Figure 28. Eclipse photography challenge Gantt chart: six concurrent challenges mapped across the eclipse timeline. The critical window around C2/C3 concentrates filter management, extreme illuminance changes, transient events, and operator cognitive overload into a narrow band where automation transitions from helpful to essential.

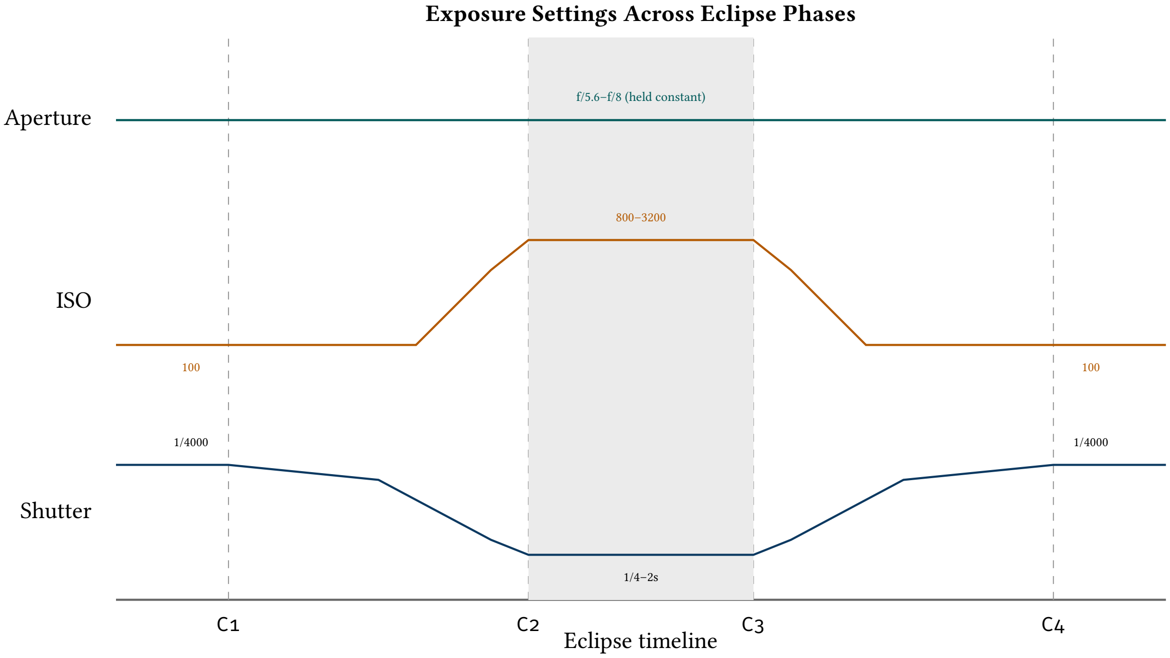


Figure 29. Multi-track exposure timeline: shutter speed and ISO ramp dramatically around C2/C3 while aperture is held constant to maintain optical quality.

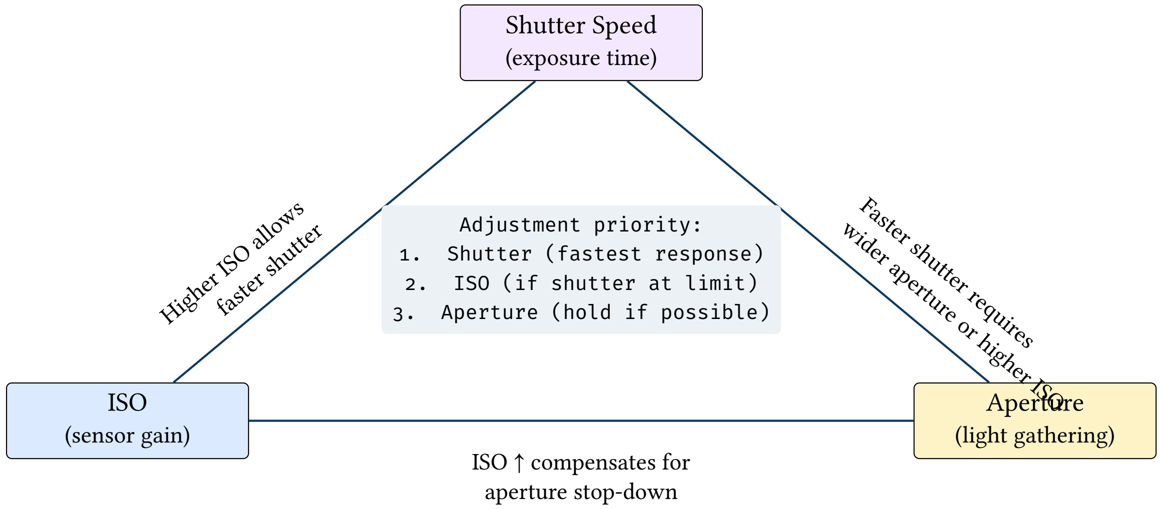


Figure 30. Exposure triangle: the three interdependent parameters. EclipsePhoto's control strategy prioritizes shutter adjustment, then ISO, while holding aperture fixed.

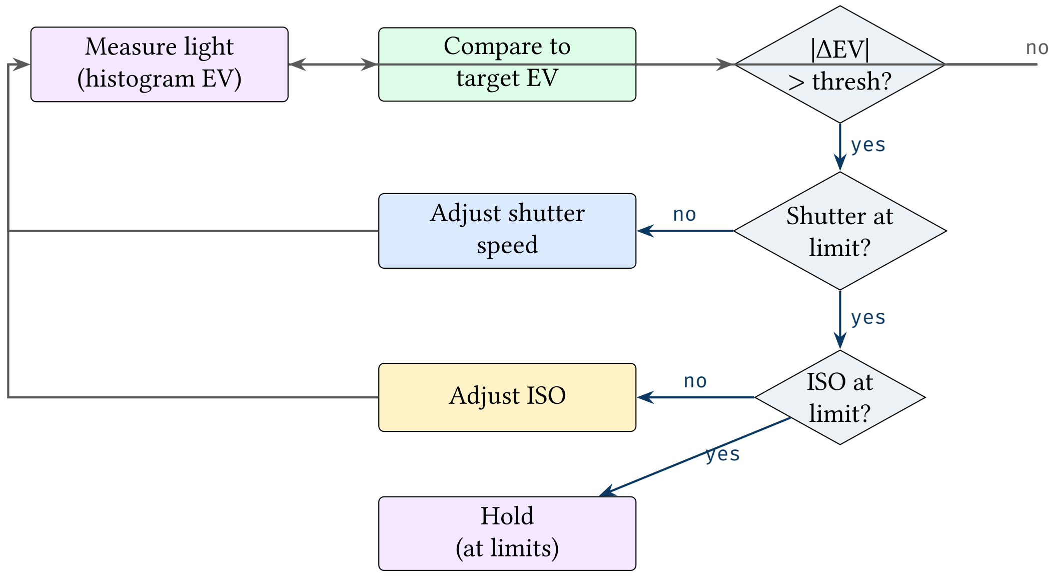


Figure 31. Automated exposure ramp strategy: measure light, compare to target, adjust shutter first, then ISO. Aperture is held constant throughout.

: Exposure Ramp Priority

Prioritize shutter speed adjustment over ISO to preserve signal-to-noise ratio; hold aperture constant to maintain optical quality and depth of field. Only increase ISO when shutter speed reaches its useful limit for the current target.

ISO Upper Limit

Set a hard ISO ceiling in EclipsePhoto's configuration—ISO 1600 for most Sony Alpha bodies, ISO 3200 at absolute maximum for the latest-generation sensors (e.g., A7R V, A1). Beyond these values, read noise dominates and fine corona structure degrades into grain. The faint outer corona and earthshine targets that demand long exposures should be reached by extending shutter speed (up to 4–8 s on a tracked mount), *not* by raising ISO. A grainy, noisy image of the corona is far worse than a slightly underexposed one—underexposure can be partially recovered in post-processing, but noise destroys the delicate radial structure that makes corona images scientifically and aesthetically valuable. EclipsePhoto must enforce this ceiling and never auto-escalate ISO beyond it, even if the histogram guard rails indicate underexposure.

8.1 Exposure Lengths at Each Stage

The following table provides worked examples of specific exposure parameters for each eclipse stage. These values assume a full-frame sensor (Sony Alpha series) and serve as starting points; EclipsePhoto's histogram guard rails will fine-tune within each stage.

Stage	Shutter	ISO	Aperture	Target
Partial (filtered)	1/1000s	100	f/8	Sunspot detail through solar filter
Diamond ring	1/4000s	100	f/5.6	Brilliant point; avoid blooming
Inner corona	1/500s	200	f/5.6	Coronal loops and base structure
Mid corona	1/60s	400	f/5.6	Streamer detail and helmet structure
Outer corona	1/2s	800	f/5.6	Faint extensions beyond 2 $R_\odot$
Earthshine	2s	1600	f/5.6	Lunar surface illuminated by Earth
Shadow bands	1/500s	1600	f/2.8	Ground pattern (wide-angle rig)

Table 4. Worked exposure examples for each eclipse stage. These are starting points; EclipsePhoto adjusts based on histogram feedback.

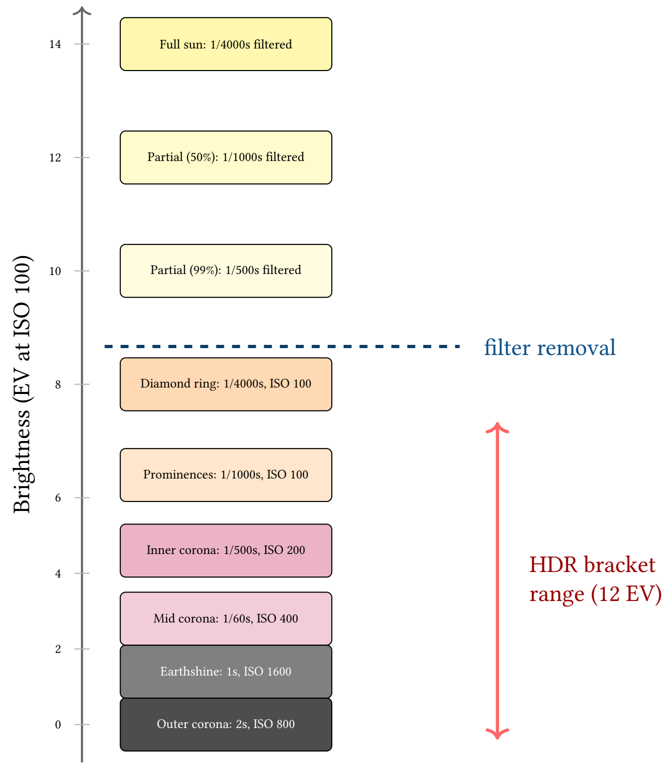


Figure 32. Exposure ladder: visual staircase showing the specific shutter/ISO combination at each eclipse stage. The 12 EV HDR bracket range during totality (red bar) spans from outer corona to prominences. The dashed line marks filter removal at C2.

Bracket Sequence Configuration. EclipsePhoto supports configurable bracket sequences to match the operator's artistic and scientific goals. The bracket parameters are:

Parameter	Default	Description
Bracket count	12 frames	Number of exposures per bracket set. More frames capture a wider dynamic range but take longer to complete.
EV step size	1 EV	Spacing between consecutive exposures. 1 EV gives good overlap for HDR merge; 2 EV covers more range but with less overlap.
Shortest exposure	1/4000s	Captures the bright inner corona and any residual Baily's beads. Limited by camera's maximum shutter speed.
Longest exposure	2–4s	Captures the faint outer corona and earthshine. Limited by tracking accuracy (untracked rigs show trailing beyond ~1s at telephoto).
Bracket direction	Short→long	Start with the shortest exposure and ramp up. This ensures the most time-critical frames (inner corona, still changing) are captured first.
Inter-frame delay	Minimum	Camera-dependent readout time. Set continuous drive mode to minimise delay between frames within a bracket.

Table 5. Bracket sequence configuration parameters. The default 12-frame, 1 EV step sequence spans ~12 EV from 1/4000s to 2s, covering the full corona dynamic range.

> **: Bracket Direction: Short to Long**
>
> Always sequence brackets from shortest to longest exposure. The inner corona and transient phenomena change fastest; capturing them first minimises motion blur and temporal misalignment within the bracket set. The faint outer corona is stable over the bracket duration (~20–30s) and can tolerate being captured last.

The operator can define multiple bracket profiles:

- **Full corona profile:** 12 frames, 1/4000s to 2s (default). Used for the primary HDR sequence during mid-totality.
- **Quick corona profile:** 7 frames, 1/4000s to 1/8s. Faster cycle time (~10s) for more bracket repetitions. Sacrifices outer corona for temporal coverage.
- **Prominence-only profile:** 5 frames, 1/2000s to 1/125s. Targeted at the inner limb region for prominence detail without wasting time on outer corona exposures.
- **Earthshine profile:** 3 frames, 1s to 4s at ISO 1600. Long exposures to capture the faintly illuminated lunar surface. Best used once during mid-totality.

EclipsePhoto can schedule a sequence of profiles across totality: for example, quick brackets during the first and last 30 seconds (when transients are most likely), full brackets during mid-totality, and a single earthshine set at the midpoint.

8.2 Aggressive Capture: Fill the Card

The bracket profiles above represent a conservative baseline. In practice, eclipses are rare, expensive to reach, and impossible to repeat—the correct strategy is to capture as much data as possible and sort it out in post-processing. Storage is cheap; missed frames are permanent.

Philosophy: Shoot Everything. A 64 GB SD card holds approximately 1,200 RAW frames at 50 MB each. A 4-minute totality at the camera's maximum continuous drive rate (10 fps on a modern mirrorless body) would produce 2,400 frames—about 120 GB. Even with a UHS-II card, the buffer will fill after 20–40 continuous frames and the camera will slow to the card's write speed. The practical limit is therefore determined by the bracket cycle time rather than the total storage capacity.

The aggressive capture philosophy is simple: *fill the card.* Take as many frames as physically possible during the critical window (C2 − 5 min to C3 + 5 min). Every frame is potentially useful for HDR stacking, noise reduction, timelapse, corona motion analysis, or creative composites. Frames that turn out to be redundant can be discarded; frames that were never taken cannot be recovered.

HDR Timelapse Strategy. The most powerful creative output from an eclipse is a high-dynamic-range timelapse video spanning the full transition from partial eclipse through totality and back. This requires continuous HDR bracket sequences from approximately 5 minutes before C2 through 5 minutes after C3—a 10+ minute window of non-stop capture.

Phase	Conservative	Aggressive (fill the card)
Partial (C1–C1+50min)	1 frame / 2 min (30 frames)	1 frame / 30s (100 frames)
Pre-ramp (C2−5min)	1 frame / 30s (10 frames)	7-frame quick bracket / 15s (140 frames)
Ramp (C2−60s)	Arm bracket, wait	7-frame bracket / 10s (42 frames)
Totality (C2–C3)	8× 12-frame bracket (96 frames)	Continuous 12-frame bracket, no pause (~16× = 192 frames)
C2/C3 transitions	2× 50-frame burst (100 frames)	2× 100-frame burst (200 frames)
Post-ramp (C3+60s)	Resume filtered	7-frame bracket / 10s (42 frames)
Post-ramp (C3+5min)	1 frame / 30s	7-frame quick bracket / 15s (140 frames)
Partial (C3+5min–C4)	1 frame / 2 min (30 frames)	1 frame / 30s (100 frames)
Total frames	**~270 frames (~14 GB)**	**~960 frames (~48 GB)**

Table 6. Conservative vs. aggressive capture strategies. The aggressive approach fills a 64 GB card and produces enough data for HDR timelapse, video stacking, and corona motion analysis.

Video Stacking for Noise Reduction. Some cameras (e.g., Sony Alpha series) can capture 4K video simultaneously with stills, or can be configured for video-only bursts. During totality, 4K video at 30 fps provides 7,200 frames across a 4-minute window. While individual video frames have lower bit depth (8-bit vs. 14-bit RAW), stacking 30–60 consecutive frames recovers dynamic range and dramatically reduces noise. The resulting "lucky imaging" composites can rival individual RAW exposures for the inner corona while providing continuous temporal coverage.

For the telephoto rigs, a hybrid approach is effective: run the HDR bracket sequence as the primary capture mode, but also record 4K video on a second card slot (if available) as insurance. The video provides continuous coverage with no gaps between bracket sets.

Corona Motion and Prominence Evolution. Continuous capture during totality enables time-series analysis that single-bracket approaches cannot:

- **Coronal streamer dynamics:** The K-corona's electron-scattered light traces magnetic field lines that evolve on timescales of minutes. A continuous bracket sequence at 30-second intervals produces enough temporal sampling to detect streamer motion and create time-lapse animations showing magnetic field evolution.
- **Prominence eruptions:** Active-region prominences can change shape visibly during a 4–7 minute totality. The mid-exposure frames (1/500–1/1000s) from successive brackets form a time series that captures loop evolution, material flows, and occasionally eruptions.
- **Shadow band propagation:** If the wide-angle rig captures shadow bands at high frame rate (60 fps video), the resulting footage can be processed to measure band spacing, velocity, and orientation—data of scientific value for atmospheric turbulence studies.

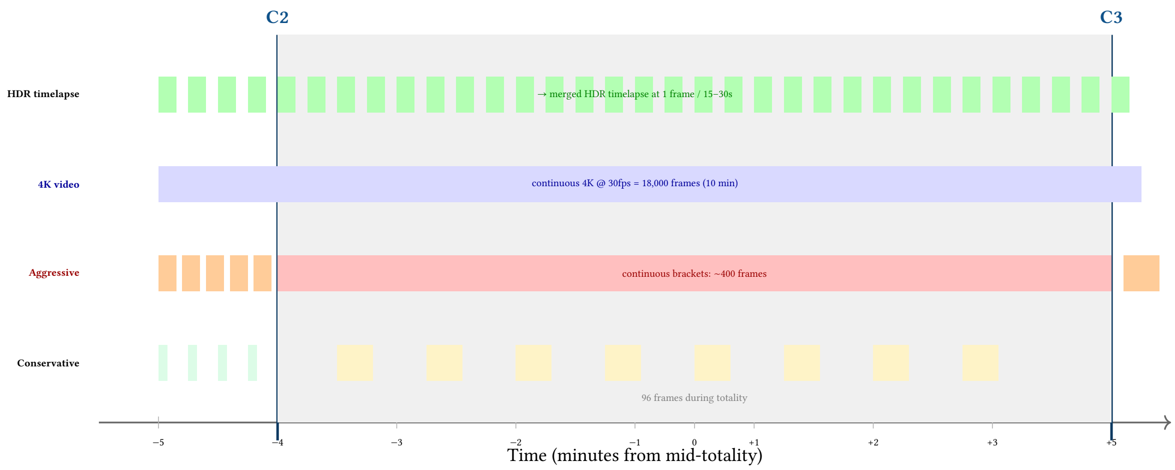


Figure 33. Capture density comparison: the conservative strategy (top) produces 8 bracket sets during totality; the aggressive strategy (middle) fills every available second with brackets and extends 5 minutes on either side of totality; 4K video (third row) provides continuous coverage for video stacking; the combined output (bottom) produces a smooth HDR timelapse spanning the full transition.

HDR Timelapse Post-Processing Pipeline. The aggressive capture strategy feeds a post-processing pipeline that produces a time-varying HDR composite:

1. **Sort and align:** Group frames by bracket set using EXIF timestamps. Align each set (sub-pixel registration for tracked rigs; solar-disc centroid alignment for untracked).
2. **HDR merge per set:** Merge each bracket set into a single 32-bit floating-point HDR frame using exposure-weighted blending (Mertens or Debevec method).
3. **Temporal smoothing:** Apply temporal median filtering across adjacent HDR frames to suppress transient noise and satellite/aircraft trails.
4. **Tone mapping sequence:** Apply consistent tone mapping (Mantiuk or Drago) across all frames to produce a viewable 8-bit timelapse sequence.
5. **Radial filtering (optional):** Apply Larson-Sekanina or unsharp-mask radial filtering to each frame to enhance coronal structure, producing a "structure timelapse" that reveals streamer motion.
6. **Encode:** Render as 4K video at 24–30 fps. A 10-minute window with 1 HDR frame every 15 seconds produces 40 frames—about 1.5 seconds of video. At 1 frame every 5 seconds (aggressive), 120 frames = 4–5 seconds of video.

> **Storage Budget for Aggressive Capture**
>
> The aggressive capture strategy generates approximately 48 GB per telephoto rig and 15 GB per wide-angle rig. Use 128 GB UHS-II cards (not 64 GB) to provide a comfortable 2.5× margin. Across a three-rig deployment, plan for 100–130 GB of total data.

: Default to Aggressive

EclipsePhoto should default to aggressive capture mode. Conservative mode exists as a fallback for constrained storage or battery situations. The operator can always discard frames in post-processing, but can never recover frames that were not captured. When in doubt, take the shot.

8.3 Focus Verification Procedures

Achieving and maintaining critical focus is essential for eclipse photography, especially at telephoto focal lengths where depth of field is measured in fractions of a millimetre. The focus verification procedure should be completed well before C1 and the focus ring locked for the duration of the eclipse.

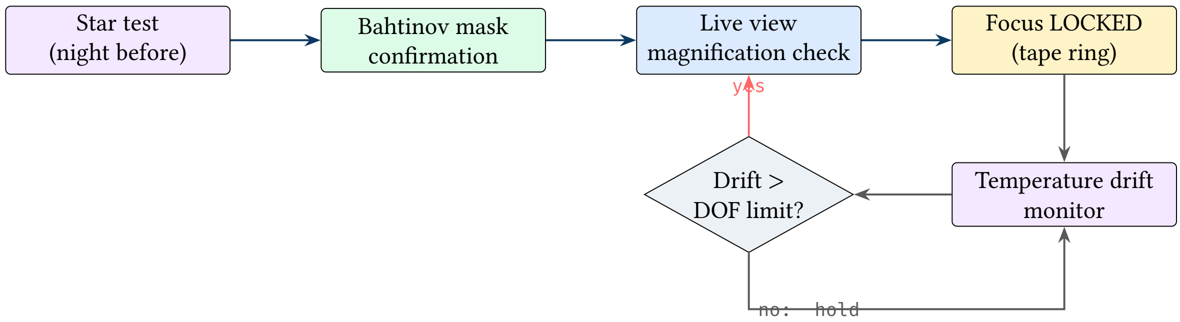


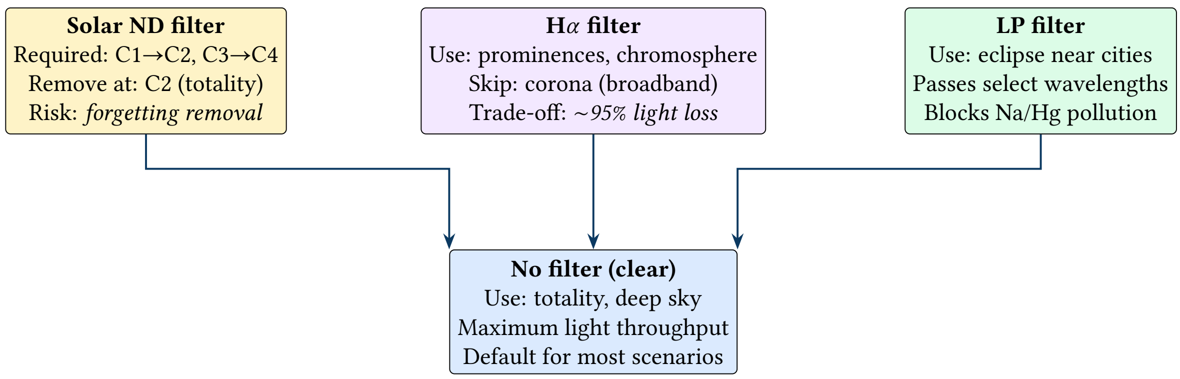
Figure 34. Focus verification procedure: star test establishes baseline, Bahtinov mask confirms critical focus, live view magnification provides final verification, then focus is locked. Temperature drift is monitored throughout the eclipse with re-check triggered if drift exceeds depth-of-field limits.

: Focus Lock Before C1

Lock focus before C1 and do not touch the focus ring during the eclipse. Any accidental bump to the focus ring during totality—when the operator is excited and potentially clumsy—will ruin every subsequent frame. Use gaffer tape to secure the focus ring after final verification.

9 Solar Filter Management

Solar filter management is the single most safety-critical aspect of eclipse photography. The filter—an optical density ~5 neutral density element—must be in place during all partial phases to protect both the camera sensor and the operator's eyes, yet must be removed precisely at C2 to allow totality capture. This section details the timing, interlocks, and automation strategy that EclipsePhoto uses to manage this critical transition (see also the state machine in Section 13).



Filter removal is a critical automated or operator action—
EclipsePhoto should track filter state and warn if inappropriate

Figure 35. Filter selection logic: each filter serves a specific phase or target. Solar filter removal at C2 is the single most time-critical manual action during an eclipse.

9.1 Filter Removal Timing

The timing of solar filter removal around C2 is the most safety-critical moment in the entire eclipse sequence. Remove too early and the camera sensor (and the operator's eyes, if looking through an optical viewfinder) is exposed to dangerous levels of unfiltered sunlight. Remove too late and the diamond ring, chromosphere, and initial corona frames are lost behind the opaque filter.

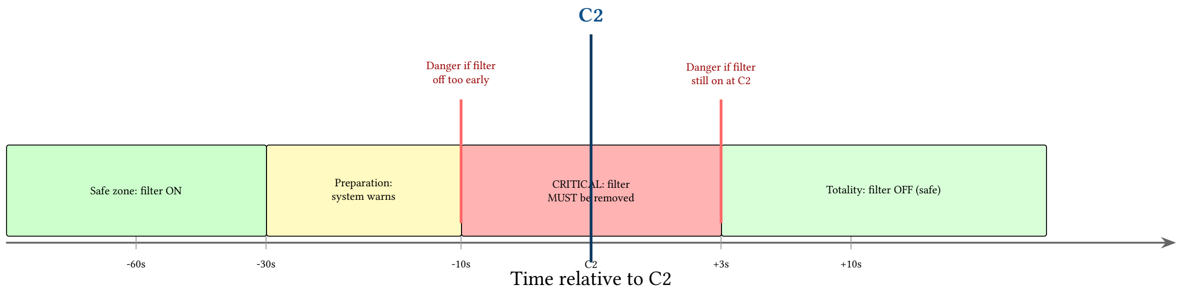


Figure 36. Filter removal timing around C2: the safe zone (green) transitions through a preparation warning zone (yellow) to the critical window (red) where the filter must be removed within seconds of C2. Removing too early risks sensor damage; too late wastes the diamond ring and chromosphere.

Sensor Damage Risk

Unfiltered sunlight through a telephoto lens can destroy a camera sensor in seconds. At 200 mm f/5.6, the lens concentrates approximately 10 W of solar energy onto the sensor—equivalent to a soldering iron. The solar filter must remain in place until the last possible moment before C2, and must be replaced immediately after C3. EclipsePhoto must track filter state and provide unmistakable audible and visual warnings at both transitions.

9.2 Safety Interlocks

EclipsePhoto implements multiple layers of safety to prevent filter-related errors:

- **Audible alarm:** A distinct audio tone begins 30 seconds before C2 ("prepare to remove filter") and changes to an urgent pattern at C2 ("remove filter NOW"). A similar sequence occurs at C3 ("replace filter NOW").
- **Full-screen warning:** The iPhone hub displays a full-screen overlay during the filter transition windows: green background when filter state matches the expected state, red background with flashing border when action is required.
- **Operator confirmation:** The system requires explicit confirmation (button press or voice command) before transitioning from filter removal to totality, and from filter replacement to partial exit. This prevents the automation from proceeding if the operator has not actually removed or replaced the physical filter.

Filter State Tracking

Filter removal and filter replacement are **operator-confirmed** states. The system should: (1) sound an audible alarm, (2) display a full-screen warning, (3) require explicit confirmation before proceeding to totality or partial exit. The system cannot physically verify filter presence—it relies on the operator to confirm.

10 Multi-Camera Deployment and Rig Strategy

An EclipsePhoto deployment typically uses multiple camera nodes for complementary coverage. The optimal geometry depends on the target: parallax-based 3D reconstruction requires baseline separation, panorama stitching requires overlapping fields, and multi-focal-length setups require co-located but differently configured nodes.

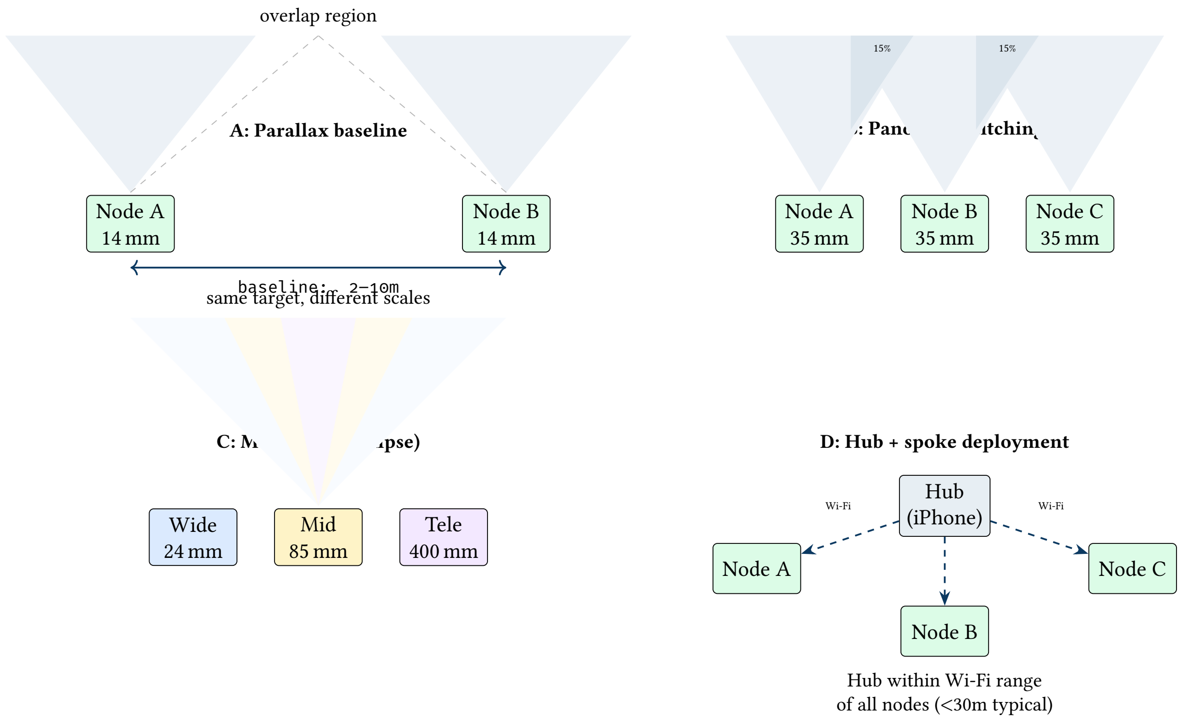


Figure 37. Multi-camera deployment geometries: (A) parallax baseline for potential 3D reconstruction, (B) panorama stitching with ~15% overlap, (C) co-located multi-focal-length for simultaneous wide/narrow coverage, (D) hub-and-spoke Wi-Fi topology constraint.

A complete eclipse imaging setup typically requires two or three rigs with different focal lengths and tracking strategies. The key insight is that the corona extends far beyond the solar disc—up to 3+ solar radii (~3° total diameter)—so a telephoto rig zoomed tightly on the Sun will miss the outer corona. Simultaneously, a wide-angle rig captures the eclipse in its landscape context for timelapse sequences.

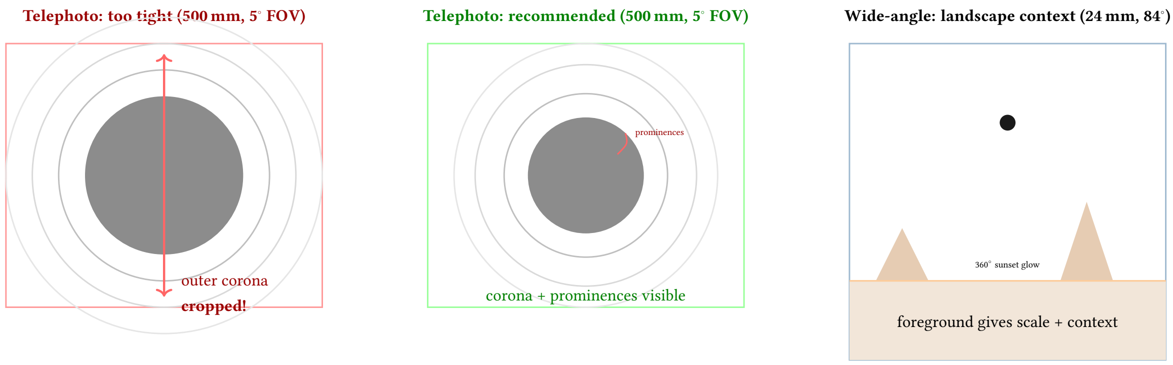


Figure 38. Corona framing: a 500 mm telephoto at very tight framing (left) risks cropping the outer corona; 500 mm with proper framing (center) captures the corona and prominences with good detail—the recommended primary corona rig; 24 mm wide-angle (right) captures the eclipse in landscape context with the 360° horizon glow visible during totality.

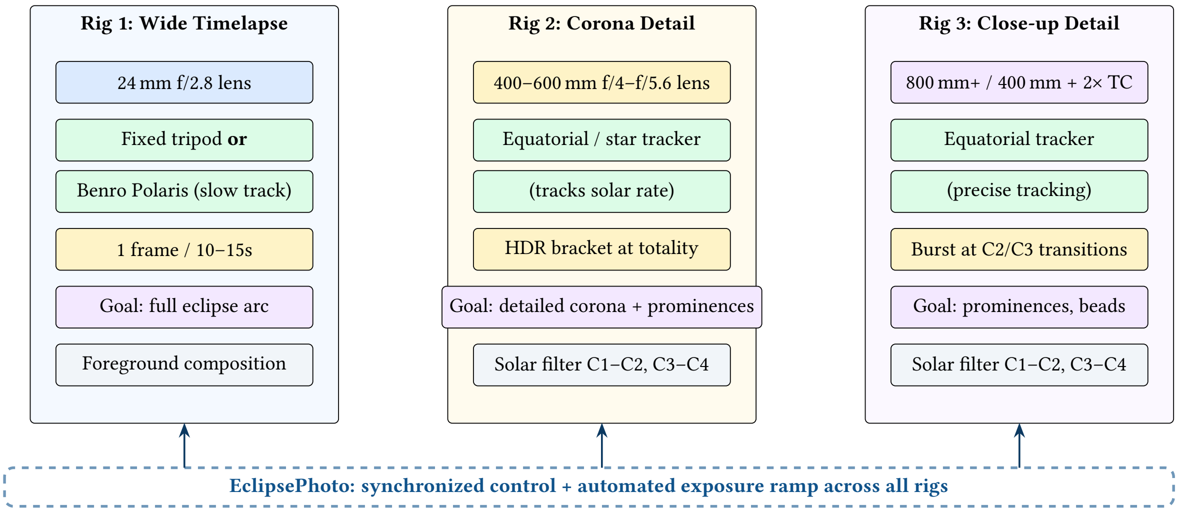


Figure 39. Three-rig eclipse deployment: (1) wide-angle on tripod or Benro Polaris for timelapse with foreground, (2) tracked 400–600 mm for detailed corona HDR with prominence detail, (3) tracked 800 mm+ for close-up prominences and transient events. EclipsePhoto coordinates exposure timing across all three.

The wide-angle rig has two operational modes:

- **Stationary (tripod):** Fixed composition with foreground (landmark, people, horizon). The Sun moves through the frame over the eclipse duration, creating a composite arc showing the progression from crescent to totality and back. Requires planning the Sun's path to ensure it stays in frame for the full 2–3 hours.
- **Slow-tracked (Benro Polaris):** The mount rotates slowly to follow the Sun, keeping it centered while the foreground gradually shifts. This produces smoother timelapse sequences and avoids the Sun drifting out of a tighter composition, but sacrifices the dramatic composite arc. Best for video-oriented output or when using moderate focal lengths (35–50 mm) where the Sun's motion would otherwise exit the frame.

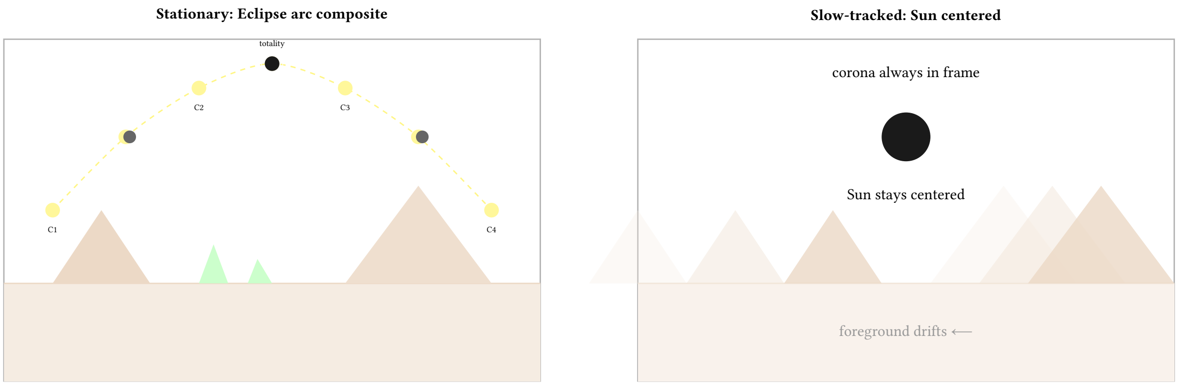


Figure 40. Wide-angle timelapse strategies: stationary (left) produces a dramatic composite arc showing the eclipse progression against fixed foreground—the classic "eclipse arc" image; slow-tracked on a Benro Polaris (right) keeps the Sun centered for smoother video timelapse but the foreground drifts. Both require planning the composition for the full eclipse duration.

Design Decision 1: Corona Framing Rule

For the primary corona rig, use 400–600 mm on full-frame (5–8° diagonal FOV). This range captures the inner and middle corona with prominence detail while fitting the outer corona to ~2.5–3 solar radii. At 500 mm (~5° FOV), the corona fills the frame productively with room for streamers. For a separate context/wide-corona rig, use 200 mm (12° FOV) to capture the full coronal extent with generous margin. When in doubt, frame wider: you can crop in post-processing but cannot recover cropped corona.

10.1 Rig Coordination via EclipsePhoto

The iPhone hub synchronizes exposure timing, filter state, and bracket sequences across all three rigs. The coordination model differs by rig role:

- **Wide rig (Rig 1):** Runs independently in timelapse mode throughout the eclipse. The hub sets the initial interval and exposure parameters but does not send per-frame commands. The wide rig does not use a solar filter (it shoots at wide angle where the Sun is a small point) and does not require filter-state synchronization.
- **Corona rig (Rig 2) and Close-up rig (Rig 3):** Receive synchronized C2/C3 commands from the hub. At C2, both rigs receive the "remove filter" warning simultaneously, followed by the "begin totality" command that starts their respective bracket or burst sequences. At C3, both receive "replace filter" in lockstep.

The hub uses GPS-synchronized time (see Section 12) to ensure that bracket start commands arrive at each node within 100 ms of each other, preventing one rig's long exposure from vibrating the tripod during another rig's short exposure (if co-mounted).

Network Architecture: Central Pi and Wi-Fi Hotspot. In a multi-rig deployment, one Raspberry Pi node is designated as the *central node*. This node runs a Wi-Fi access point (hostapd) that creates a dedicated EclipsePhoto network, isolated from any public or venue Wi-Fi. The other Pi nodes connect as clients. The iPhone connects to the same access point as a dashboard and control interface—it does not need to act as the network hub itself.

This architecture provides several advantages over using the iPhone as the network center:

- **Reliability:** The Pi's Wi-Fi is dedicated to EclipsePhoto; the iPhone may receive calls, notifications, or iOS updates that disrupt connectivity.
- **GPS time source:** The central Pi serves as the NTP time source for all other nodes and the iPhone, ensuring all clocks are synchronized to GPS (see Section 18).
- **Autonomous operation:** If the iPhone disconnects (battery, crash, pocket mode), the Pi nodes continue operating autonomously—the hub commands are optimizations, not requirements. Each node carries its own ephemeris and can execute the full eclipse sequence independently.
- **Safety margin:** A dedicated Wi-Fi network avoids contention with other devices at crowded eclipse-viewing sites.

> **: Autonomous Node Principle**
>
> Every Pi capture node must be capable of executing the complete eclipse sequence—from C1 through C4—without any hub communication. The iPhone dashboard and central Pi coordination improve timing precision and operator awareness, but the system must degrade gracefully to fully autonomous per-node operation if the network fails.

Supplementary Video: GoPro and Action Cameras. A GoPro Hero (or similar action camera) can serve as a simple, self-contained supplementary capture device alongside the primary Sony rigs. Mounted overhead on a clamp or positioned on a secondary tripod pointed at the Sun, the camera records continuous 4K or 5.3K video throughout the entire eclipse—started before C1 and stopped after C4—with zero operator intervention required during the event.

The benefits of this approach are significant. First, the action camera captures the entire eclipse as a single, unbroken video file, providing an insurance policy against automation failures on the primary rigs. Second, the recorded video serves multiple post-processing workflows: it can be extracted into timelapse frames at arbitrary intervals, it enables video stacking (analogous to the lucky-imaging technique used in planetary astrophotography, where thousands of frames are registered and the sharpest percentile combined), and the audio track preserves crowd and observer reactions—an often-overlooked but irreplaceable record of the experience.

The wide-angle lens characteristic of action cameras is an advantage here, not a limitation. The expansive field of view naturally captures the 360° horizon glow during totality, the approaching umbral shadow sweeping across the landscape, and the dramatic darkening of the sky—phenomena that a telephoto rig pointed at the Sun cannot record. This wide-field perspective complements the narrow-field corona and close-up data from the primary rigs.

Modern GoPro cameras support LOG and flat colour profiles (e.g., GoPro's GP-Log or Protune Flat) that preserve additional dynamic range in the recorded video, providing greater latitude in post-processing for scenes with extreme contrast such as the diamond ring against the darkened sky. Some models also support RAW-format frame grabs and GPS metadata embedding, which can assist with time alignment against the primary rig data.

A GoPro is not a replacement for the primary Sony rigs—it lacks the sensor size, bit depth, and telephoto reach needed for scientific-quality corona imaging—but it is a valuable supplementary data source that requires minimal setup and provides high-value footage with essentially no risk of operator error.

10.2 Site Scouting and Equipment Setup

Arrive at the observing site at least one day before the eclipse. The scouting visit serves several purposes: confirm an unobstructed view along the solar bearing at the time of totality, identify the best ground for tripod stability, check for potential vibration sources (roads, generators, crowds), and plan cable routing between rigs. Photograph the site during the corresponding time window so that shadows, foreground options, and potential obstructions are known before eclipse morning.

On eclipse morning, deploy equipment at least 3 hours before C1. This allows time for thermal equilibration (lenses and camera bodies expand or contract as they warm in direct sunlight, shifting focus), GPS cold-start acquisition (Section 18), and a complete dry run of the automation sequence (Section 16).

> **: Night-Before Star Alignment**
>
> If the site permits overnight access, set up tracked rigs the night before and perform polar alignment using stars. This is by far the most accurate method—the Polaris-based drift alignment or plate-solving routine achieves arc-minute accuracy that is difficult to replicate during daytime. Mark tripod leg positions with tape or paint so the rig can be removed for transport and returned to exactly the same orientation.

10.3 Mount Types and Tracking Options

Different rigs benefit from different tracking strategies:

- **No tracking (fixed tripod):** Suitable for the wide-angle rig at short focal lengths (24–35 mm). The Sun drifts by approximately 15 arcseconds per second, which is negligible across a wide field of view. A fixed tripod also produces the cleanest foreground for composite arc images. At 24 mm on a full-frame sensor, the Sun takes over 3 hours to cross the frame.
- **Benro Polaris (or similar smart tripod head):** A motorised pan-tilt head that can track celestial objects at sidereal or solar rate. The Benro Polaris is compact, runs from an internal battery, and can be set up without polar alignment—it uses an internal IMU and GPS to compute tracking motion. Ideal for the wide rig at moderate focal lengths (35–85 mm) where fixed-tripod drift would push the Sun out of frame during the eclipse. Tracking accuracy is lower than a true equatorial mount (~1–2 arcminutes over 30 minutes), which is adequate for wide-angle work but not for telephoto.
- **Star tracker (iOptron SkyGuider Pro, Sky-Watcher Star Adventurer):** Compact equatorial mounts designed for camera lenses. These clamp between a ball head and tripod, and rotate a single axis at sidereal rate (adjustable for solar rate). Polar alignment is required—typically via a built-in polar scope sighted on Polaris. Tracking accuracy of ~10–30 arcseconds over several minutes makes these suitable for the 400–600 mm corona rig, where untracked exposures beyond ~1 second show trailing.
- **Equatorial telescope mount (iOptron CEM26, Sky-Watcher EQ6-R):** Full equatorial mounts with motorised tracking in both axes and periodic error correction. These provide the highest tracking accuracy (<5 arcseconds over several minutes) and are necessary for the 800 mm+ close-up rig, where any tracking error is magnified to visible trailing. Heavier and more complex to set up, but essential for long-exposure outer corona capture at high magnification.

10.4 Daytime Alignment Procedures

If night-before alignment is not possible, several daytime methods are available:

- **Solar alignment:** Point the mount at the Sun (using a solar filter or the camera's shadow on the ground) and verify that tracking keeps the Sun centred over 5–10 minutes. Adjust the mount's polar axis until drift is minimised. This iterative process is less precise than star alignment but achievable in 15–20 minutes. Most star tracker apps (e.g., iOptron Commander, SynScan) include a solar tracking mode for this purpose.
- **Compass and inclinometer:** Set the mount's polar axis to the local latitude using an inclinometer, and point it north (or south in the southern hemisphere) using a compass corrected for local magnetic declination. This gets within 1–2° of true polar alignment—adequate for focal lengths below 300 mm but marginal for longer lenses.
- **Daytime plate solving:** Some mount controllers (and apps like ASTAP or All-Sky Plate Solver) can plate-solve on the Sun's position using a solar filter image, computing the mount's pointing error and correcting it. This automates the iterative solar alignment process and can achieve sub-arcminute accuracy if the software supports solar disc detection.

- **Moon alignment (if visible):** If the Moon is visible during the day (common in the days around a solar eclipse, since the Moon is near the Sun), it can serve as an alignment target. The Moon's sharp limb and surface features provide precise feedback on tracking accuracy at telephoto focal lengths.

Thermal Focus Drift

After daytime alignment, the rig will be in direct sunlight for hours. Metal lens barrels expand, shifting the focus point. Carbon fibre tubes are more thermally stable. Regardless of material, re-check focus every 30 minutes during the partial phase (see Section 15) and immediately after removing the solar filter at C2.

11 HDR Compositing Pipeline

The 12+ EV dynamic range of the solar corona requires compositing multiple exposures. The processing pipeline aligns frames, weights them by exposure, and applies specialized radial filtering to reveal coronal structure that no single exposure can capture (see Section 5).

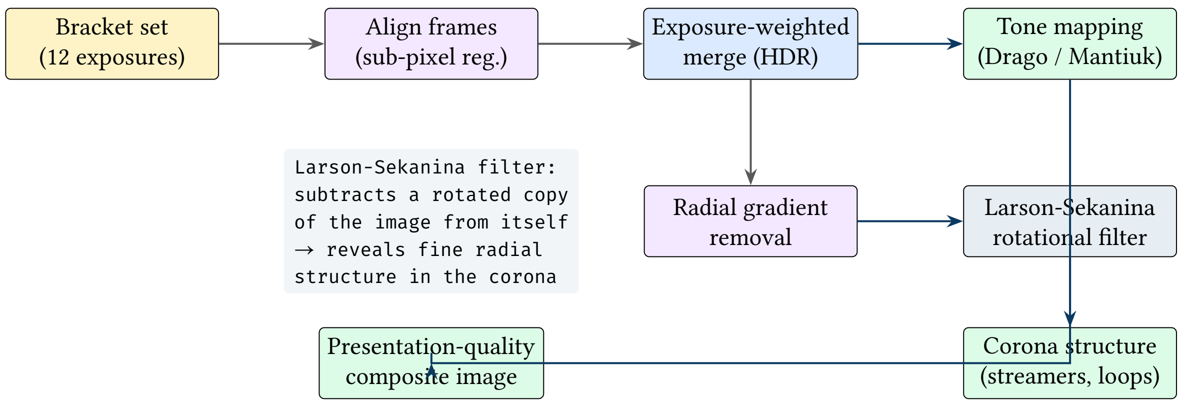


Figure 41. HDR compositing pipeline for corona imagery: bracket exposures are aligned, merged with exposure-based weighting, and processed through two paths—tone mapping for natural appearance, and radial filtering (Larson-Sekanina) to reveal fine coronal structure.

Noise Reduction via Stacking. During totality, multiple bracket sets are captured in sequence. If totality lasts 4 minutes and a full 12-exposure bracket takes approximately 30 seconds (including readout and settling time), approximately 8 bracket sets can be acquired. Each bracket set contributes one frame at each exposure level.

For the faintest exposures (outer corona at 1–2 seconds), individual frames are dominated by read noise and photon shot noise. Stacking N aligned frames from successive bracket sets improves the signal-to-noise ratio by a factor of $\sqrt{N}$. With 8 bracket sets, stacking the outer-corona frames yields $\sqrt{8} \approx 2.8\times$ SNR improvement—equivalent to roughly 1.5 stops of additional effective dynamic range at the faint end.

The alignment step is straightforward for tracked rigs (the Sun remains stationary in the frame), but for untracked rigs the solar disc drifts by approximately 15 arcseconds per second. Over a 4-minute totality, this amounts to $\sim 1^{\circ}$ of drift—significant at telephoto focal lengths. EclipsePhoto should record timestamps with each frame to enable precise alignment during post-processing.

12 Pre-Totality Operational Timeline

An eclipse totality sequence requires precisely timed actions. This operational countdown ties the physical phenomena (contact points, illuminance changes) to the concrete camera operations that EclipsePhoto must automate or prompt.

Time	Action
T−24h	Check solar activity (SDO, SWPC); note active regions for prominence targets
T−4h	Equipment setup; GPS sync verification; test exposures on daytime sky
T−2h	Star focus calibration (if pre-dawn setup); Bahtinov mask on bright star
T−1h	Solar filter on all rigs; frame composition; test filtered exposure on Sun
T−60min	C1: begin partial-phase capture (1 frame / 2 min)
T−30min	Verify tracking; check focus has not drifted from thermal changes
T−10min	Temperature drop check; arm bracket sequence on corona and close-up rigs
T−5min	Final focus verification via live view magnification; prepare totality exposure table
T−30s	Shadow bands may appear; switch wide rig to video mode if desired
T=0 (C2)	**REMOVE SOLAR FILTER**; start HDR bracket on all telephoto rigs
T+mid	Execute brackets continuously; check histogram after each set; adjust if needed
T=end (C3)	**REPLACE SOLAR FILTER**; resume partial-phase capture
T+60min	C4: end of eclipse; stop all capture sequences

Table 7. Expanded operational timeline from T−24h through C4. Bold entries are safety-critical filter actions.

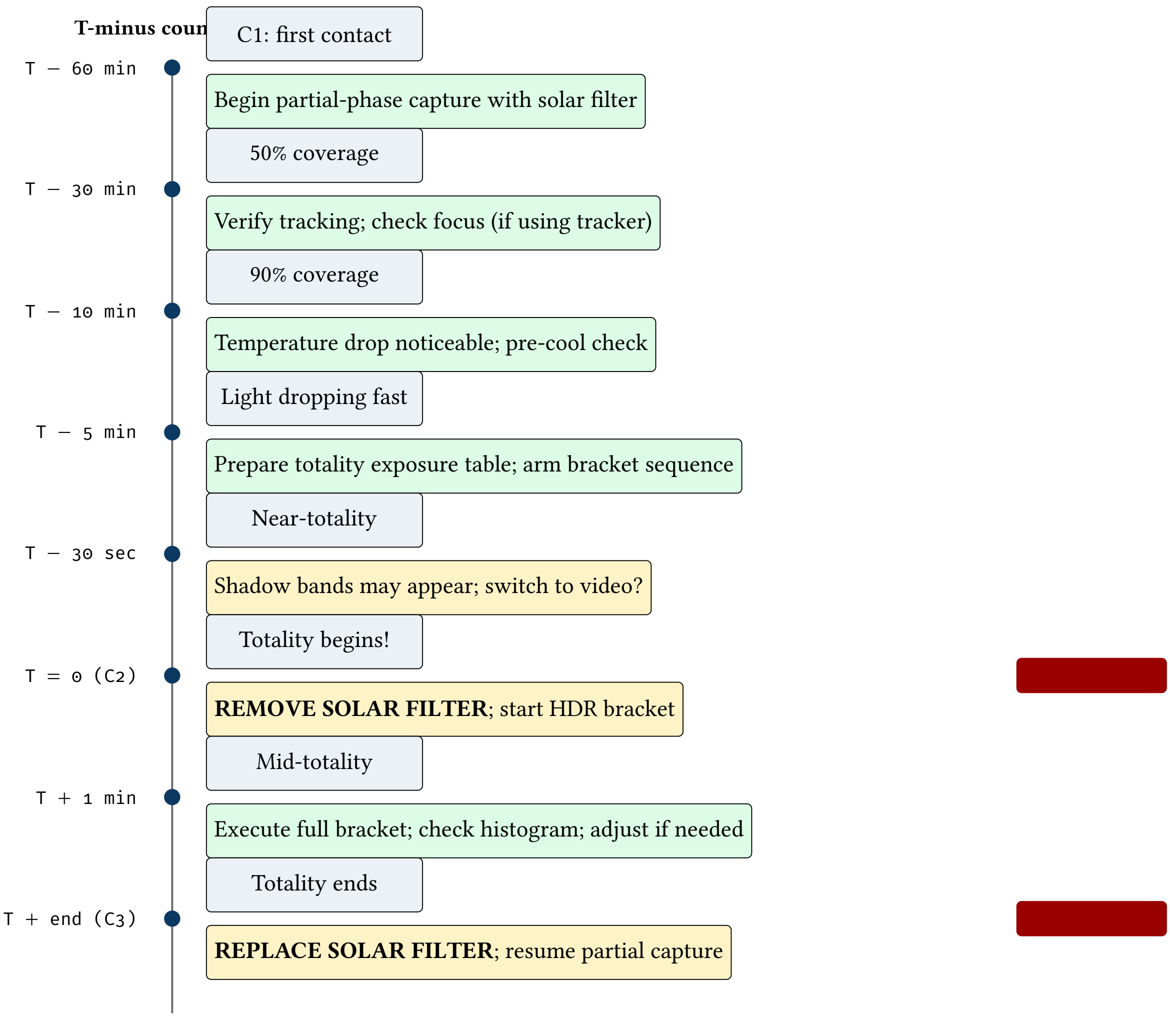


Figure 42. Pre-totality operational countdown: timed sequence of camera actions keyed to eclipse contact points. Solar filter removal at C2 and replacement at C3 are the most safety-critical actions (both for equipment and for image capture).

Solar Filter Safety

Failure to remove the solar filter at C2 wastes the entire totality window (the filter blocks >99.999% of light). Failure to replace the filter at C3 risks sensor damage from unfiltered sunlight. EclipsePhoto should track filter state and issue prominent warnings at C2 and C3 transitions.

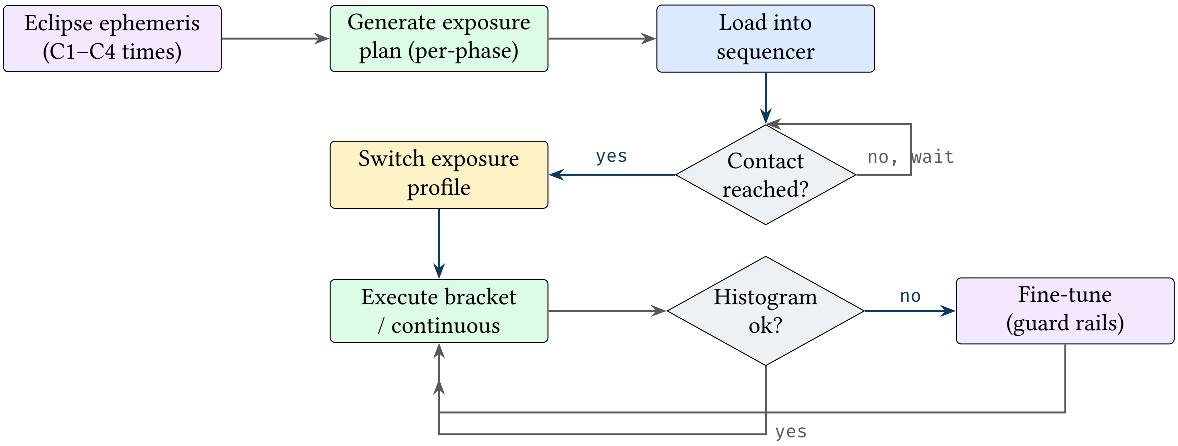


Figure 43. Eclipse automation pipeline: pre-computed ephemeris drives a time-keyed exposure plan, with histogram-based guard rails providing fine-tuning within each phase. This combines the predictable timing of eclipses with reactive histogram-based guard rails.

12.1 Totality Countdown Clock

During totality itself, every second counts. The following diagram shows the second-by-second action plan for a typical 4-minute totality window. EclipsePhoto executes this sequence autonomously, freeing the operator to observe the eclipse visually.

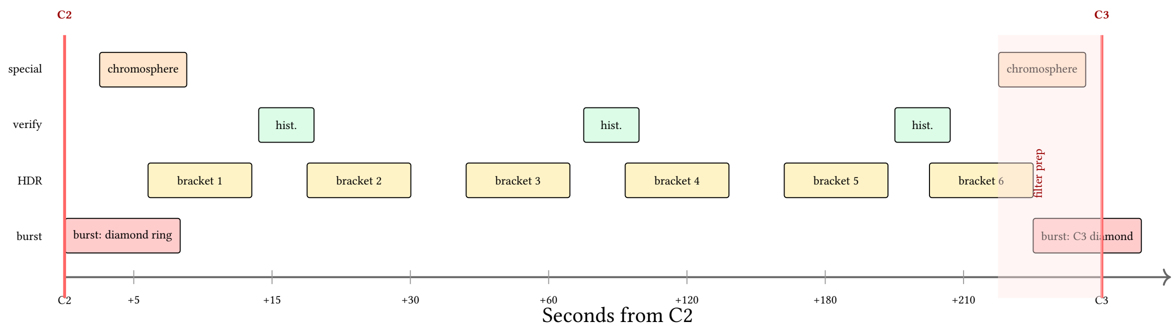
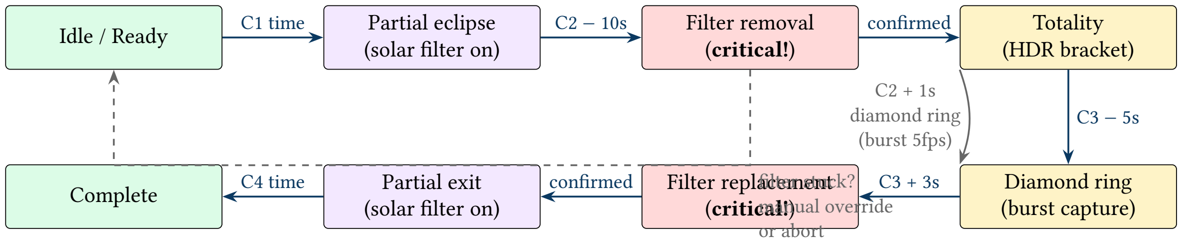


Figure 44. Totality countdown clock: second-by-second action plan for a ~4-minute totality. Burst capture at C2/C3 transitions, HDR brackets every ~30 seconds, histogram verification between brackets, and chromosphere capture at the edges. The red zone near C3 indicates the filter replacement preparation window.

13 Eclipse State Machine

The capture node state machine defined in the system architecture can be extended with eclipse-specific states that map to the physical phenomena described in Sections 2–6. This bridges the gap between the astronomical science and the system architecture.



Filter removal and filter replacement are **operator-confirmed** states.
The system should: (1) sound an audible alarm, (2) display a full-screen warning,
(3) require explicit confirmation before proceeding to totality or partial exit.

Figure 45. Eclipse-specific state machine: extends the capture node states with eclipse phases. Filter removal (at C2) and filter replacement (at C3) are operator-confirmed critical transitions. The diamond ring state triggers burst capture at the C2/C3 transitions.

14 Telemetry Dashboard

During an eclipse, the iPhone hub must display time-critical information at a glance. This mockup shows the essential telemetry elements for eclipse operations, organized by urgency.

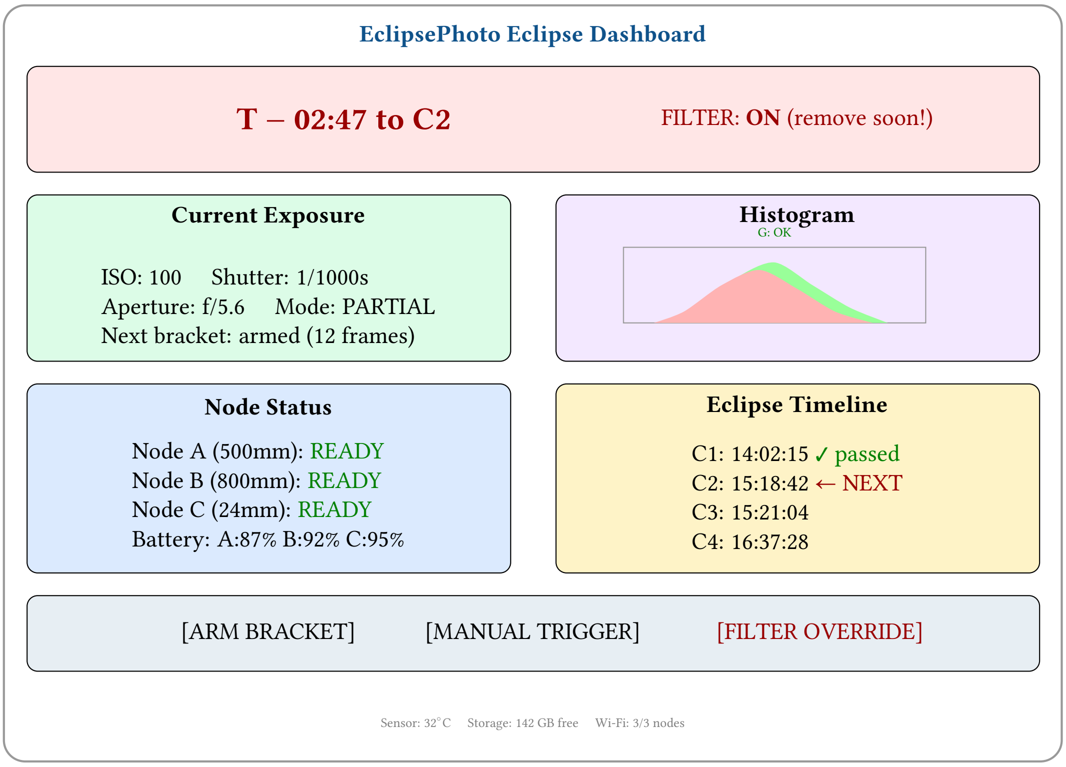


Figure 46. Eclipse telemetry dashboard mockup: the countdown to the next contact point dominates the top bar; filter state is always visible with prominent warnings. Current exposure, live histogram, node status, and eclipse timeline are shown in a four-panel layout with action buttons for manual override.

Design Decision 2: Dashboard Design: Eclipse Mode

The eclipse dashboard prioritizes three things above all else: (1) countdown to next contact point, (2) solar filter state with audible/visual warnings, and (3) bracket sequence readiness. All other telemetry (node health, storage, temperature) is secondary. The filter state indicator must be the largest, most prominent element on screen during the C2 and C3 transitions.

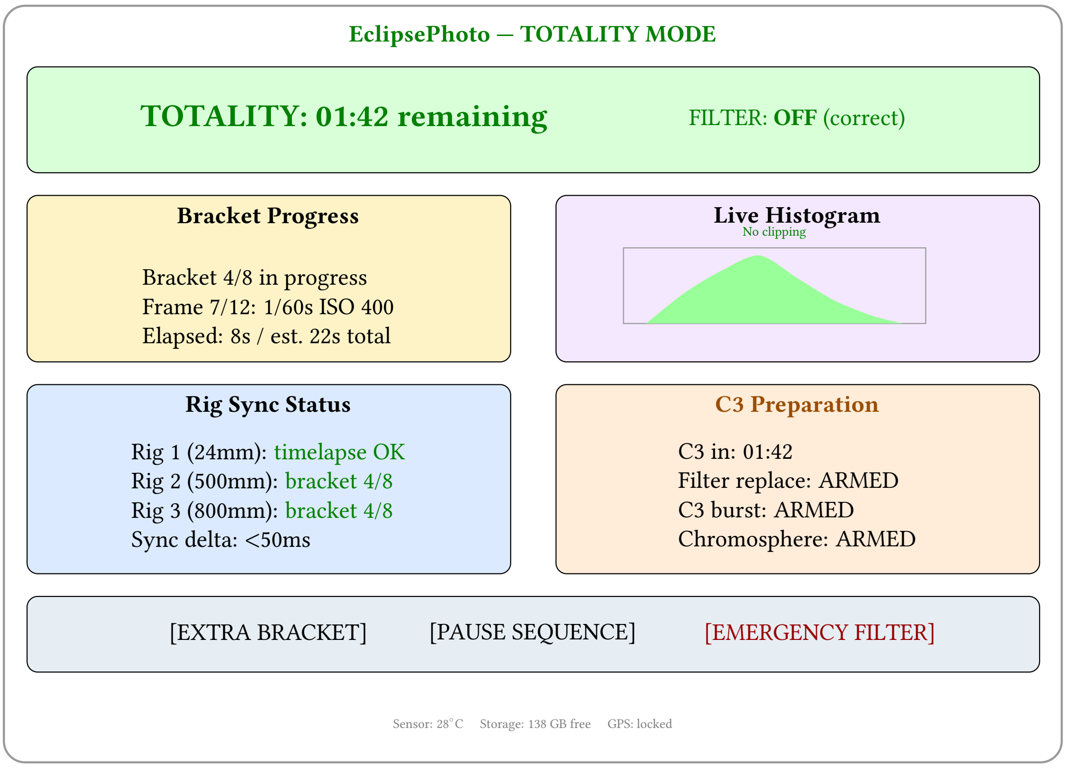


Figure 47. Totality-mode dashboard: during totality the colour scheme inverts to green (filter correctly removed), bracket progress replaces exposure settings, C3 preparation status shows readiness for the exit transition. The "emergency filter" button provides a manual override if conditions change unexpectedly.

15 Focus and Optical Quality

15.1 Star Focus Calibration

Critical focus at telephoto focal lengths requires calibration on a point source—ideally a bright star. The night before the eclipse (or in the pre-dawn hours if the setup is done early), point the telephoto rig at a bright star (Vega, Arcturus, or Sirius work well) and use live view at maximum magnification to achieve the smallest possible point spread function.

The star test procedure is straightforward:

1. Point the telescope or telephoto lens at a bright star.
2. Switch to manual focus and enable live view at maximum magnification (10× or higher).
3. Slowly adjust the focus ring until the star image is the smallest, sharpest point. Overshoot in both directions to confirm you have found the true minimum.
4. If the camera supports focus peaking, enable it as a secondary confirmation.
5. Lock the focus ring with gaffer tape.

If the eclipse occurs during the day and no pre-dawn setup is possible, focus can be achieved on the solar disc itself through the solar filter, using sunspot edges or the solar limb as focus targets. This is less precise than a star test but adequate for most focal lengths below 400 mm.

15.2 Software-Assisted Focus

EclipsePhoto provides real-time focus assistance through the Raspberry Pi's live view feed, going beyond the camera's built-in focus aids. The system captures a continuous stream of preview frames and provides computed focus metrics to the operator via the iPhone dashboard.

Live Sharpness Meter. The Pi continuously computes a sharpness metric (Laplacian variance) on a user-selectable region of interest (ROI) in the live view feed. This metric is displayed as a real-time bar graph on the iPhone dashboard, allowing the operator to "peak" the focus by slowly rotating the focus ring and watching the meter climb. The peak of the sharpness curve corresponds to best focus.

Unlike the camera's built-in focus peaking (which highlights edges with a colour overlay), the computed sharpness meter provides a *quantitative* readout that makes it easy to detect sub-pixel focus improvements. The operator can see whether a 1° rotation of the focus ring improved or degraded sharpness, even when the change is invisible on the camera's rear LCD.

Focus Bracketing for Calibration. For initial focus calibration (night before or pre-dawn), EclipsePhoto can automate a focus bracket: the system drives a motorized focuser (e.g., ZWO EAF or similar stepper-based unit) through a range of positions, capturing a frame at each step and computing the sharpness metric. The position with the highest sharpness is identified as best focus, and the focuser is driven to that position.

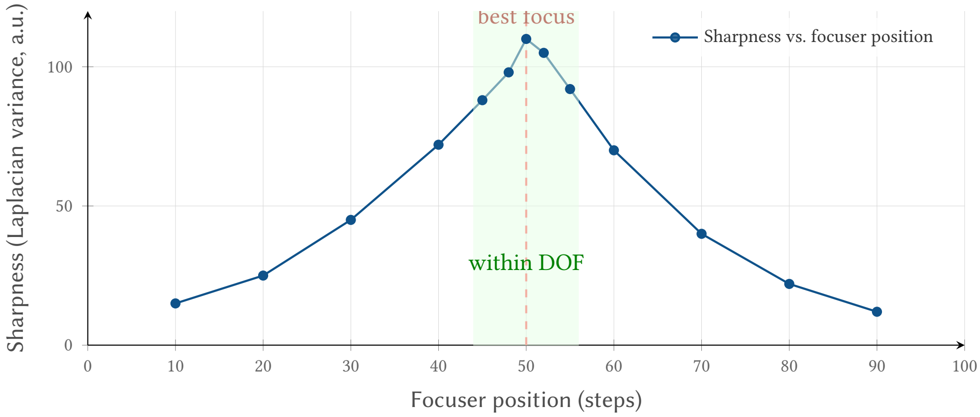


Figure 48. Automated focus calibration: the focuser sweeps through a range of positions while EclipsePhoto measures sharpness at each step. The peak (dashed red line) identifies best focus; the green zone indicates positions within the depth of field.

This automated calibration eliminates the subjectivity of manual focus adjustment and provides a repeatable, quantitative result. The calibration curve also reveals the depth of field directly: the width of the peak at (for example) 80% of maximum corresponds to the usable DOF range.

Continuous Focus Monitoring. Once focus is locked, EclipsePhoto continues to compute the sharpness metric on every captured frame during the eclipse. The dashboard displays:

- **Current sharpness:** real-time bar, colour-coded green (within 90% of baseline), amber (70–90%), red (<70%).
- **Sharpness trend:** a sparkline graph of the last 20 measurements, making drift immediately visible.
- **Temperature correlation:** ambient temperature plotted alongside sharpness trend, showing whether focus changes track thermal drift.

If sharpness drops below the amber threshold and a motorized focuser is connected, EclipsePhoto can optionally apply a micro-correction between bracket sets—nudging the focuser by a computed number of steps based on the temperature-focus calibration curve established during the pre-eclipse calibration sweep.

Motorized Focus During Totality

Automated focus correction during totality is a double-edged sword: a successful correction saves the remaining frames, but an overcorrection (due to noise in the sharpness metric or vibration from the focuser motor) could make things worse. Enable auto-correction only if (a) a motorized focuser is fitted, (b) the pre-eclipse calibration curve is available, and (c) the operator has tested the correction loop during the dry run. When in doubt, leave focus locked and accept mild softness—slightly soft data is recoverable; badly overcorrected data is not.

15.3 Temperature-Induced Focus Drift

Telescope and telephoto lens barrels are typically constructed from aluminium alloys, which expand and contract with temperature changes. During an eclipse, the ambient temperature can drop by 3–8°C in the final minutes before totality as the Sun's energy input decreases. This thermal contraction shifts the focal plane.

At 200 mm f/4, a 10°C temperature drop can shift focus by approximately 50–100 μm. For context, the depth of field at f/5.6 and 200 mm focused at infinity is roughly 200 μm—so a 10°C swing can consume half the available depth of field.

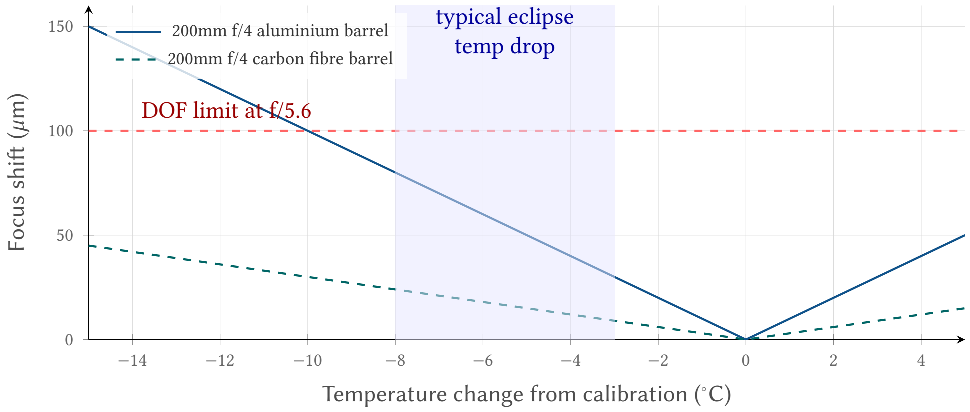


Figure 49. Focus shift vs. temperature change for typical telephoto lenses. The blue shaded region shows the 3–8°C temperature drop typical during the final partial phases of an eclipse. Aluminium-barrel lenses can shift by 30–80 μm in this range, approaching the depth-of-field limit at f/5.6.

The system should log ambient temperature (from the Raspberry Pi's onboard sensor or an external probe) and warn if the cumulative temperature change since focus calibration exceeds a configurable threshold (default: 5°C). If pre-compensation is desired, the operator can intentionally defocus slightly in the direction of expected drift.

15.4 Bahtinov Mask Procedure

A Bahtinov mask is an optical device placed over the lens aperture that creates a distinctive three-spike diffraction pattern from point sources. Perfect focus is achieved when the central spike exactly bisects the angle formed by the other two spikes. Any defocus causes the central spike to shift laterally, providing a clear and unambiguous visual indicator.

The Bahtinov mask is the gold standard for focus confirmation because it removes the subjectivity of judging "minimum star size" by eye. For telephoto lenses, commercial Bahtinov masks are available in standard filter thread sizes; for telescopes, they can be 3D-printed or laser-cut to fit the aperture.

15.5 Lens Breathing

Some lenses—particularly cinema-style lenses and certain older designs—exhibit "focus breathing": a slight change in field of view when the focus position changes. This means that after achieving perfect focus, the framing may have shifted slightly compared to the initial composition.

Always verify framing *after* the final focus adjustment, not before. If the lens breathes significantly (more than 2–3% FOV change), re-compose after locking focus and note that the framing will be stable from that point forward.

15.6 Automated Focus Quality Detection

While mechanical focus adjustment during the eclipse is impractical (risk of overshoot, vibration, and lost frames), EclipsePhoto can *detect* focus degradation in real time by analyzing captured frames. If focus loss is detected, the system can alert the operator rather than silently capturing soft images for the remainder of totality.

The detection strategy uses the sharpest frames from each HDR bracket (typically the mid-exposure frames of the inner corona, which contain high-contrast radial structure):

1. **Edge contrast metric:** Compute the Laplacian variance (or Tenengrad gradient magnitude) of a central crop around the solar limb. A sharp image produces a high Laplacian variance; a defocused image produces a lower value.
2. **Baseline comparison:** The first bracket's sharpness metric establishes the baseline. Subsequent brackets are compared: if the metric drops below 70% of baseline, the system flags a "focus degradation" warning.
3. **Trend detection:** A monotonically decreasing sharpness trend across 3+ consecutive brackets strongly suggests thermal focus drift rather than atmospheric seeing (which would produce random variation).

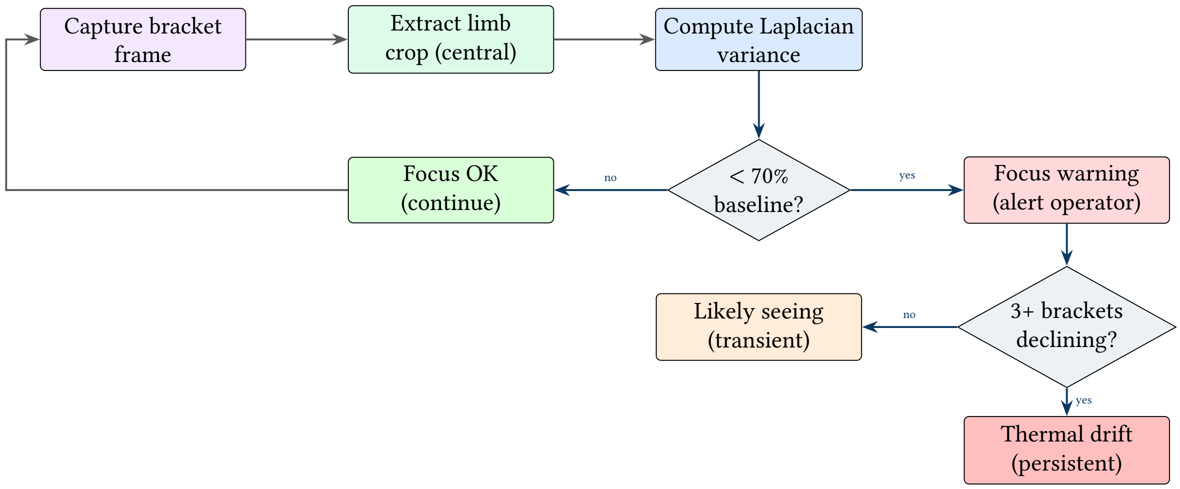


Figure 50. Focus quality detection pipeline: each bracket frame's limb region is analyzed for edge sharpness using Laplacian variance. A drop below 70% of baseline triggers a warning; a sustained declining trend across multiple brackets indicates thermal focus drift rather than atmospheric seeing.

If thermal drift is confirmed, the operator has limited options mid-totality: stopping to refocus risks losing 30+ seconds of bracket time and introduces vibration. The recommended response depends on the severity:

- **Mild (<30% degradation):** Continue shooting. The softness may be recoverable with deconvolution sharpening in post-processing, and the data is still scientifically useful.
- **Severe (>50% degradation):** If using a motorized focuser (e.g., ZWO EAF), EclipsePhoto can attempt a micro-adjustment of 10–20 μm between brackets. Without a motorized focuser, alert the operator via the dashboard and let them decide.

> **Focus Detection Limitations**
>
> Focus quality detection requires high-contrast features in the frame. During the very shortest exposures (outer corona), the signal-to-noise ratio may be too low for reliable sharpness measurement. Use the mid-exposure frames (inner corona, 1/60–1/500s) for focus assessment, as they contain well-exposed radial structure with clear edges.

> **: Focus Lock Protocol**
>
> Lock focus before C1. Do not adjust focus during the eclipse unless automated detection confirms severe drift *and* a motorized focuser is available. If temperature drift is expected, pre-compensate based on the forecast temperature at mid-totality. Verify framing after locking focus to account for any lens breathing.

16 Robustness and Sanity Checks

16.1 Pre-Event Dry Run

On the day before the eclipse (or earlier), run the full EclipsePhoto automation sequence with simulated contact times set 5 minutes apart. This dry run should exercise every state transition:

1. Idle → Partial eclipse: verify the intervalometer starts and captures filtered frames.
2. Partial eclipse → Filter removal: verify the audible alarm sounds and the full-screen warning appears.
3. Filter removal → Totality: verify the bracket sequence starts after operator confirmation.
4. Totality → Diamond ring: verify the burst capture triggers at the correct time offset.
5. Diamond ring → Filter replacement: verify the replacement warning sounds.
6. Filter replacement → Partial exit: verify filtered capture resumes.
7. Partial exit → Complete: verify clean shutdown.

Any failures in the dry run must be resolved before the actual eclipse. There is no opportunity to debug during totality.

16.2 Exposure Verification

After each exposure, EclipsePhoto should perform a rapid histogram analysis to verify that the image is neither clipped nor underexposed. The guard rails are:

- **Highlight clipping:** If the 99.5th percentile of pixel values exceeds 252/255 (in 8-bit representation), the frame is considered clipped. Reduce exposure by 1 stop for the next frame and flag the current frame.
- **Shadow noise floor:** If the 0.5th percentile of pixel values is below 3/255, the frame may be underexposed (or the scene is simply very dark, as expected for outer corona frames). If the current stage expects bright content (inner corona, diamond ring), increase exposure by 1 stop.

These guard rails operate within each stage's expected range and do not override the bracket sequence—they adjust the baseline around which the bracket is centered.

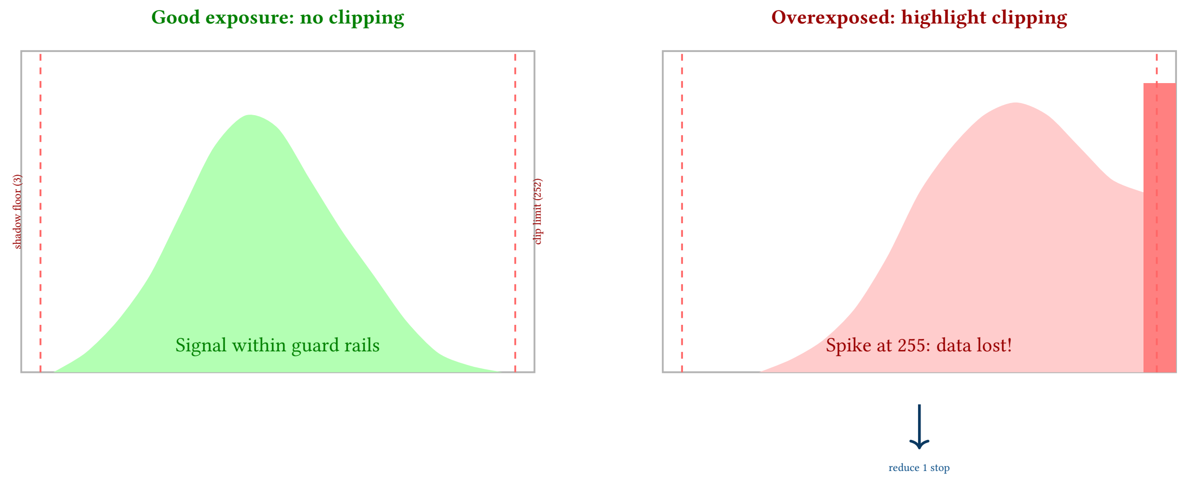


Figure 51. Histogram guard rails: a well-exposed frame (left) has signal between the shadow floor (3/255) and clip limit (252/255). A clipped frame (right) shows a spike at 255—irrecoverable data loss in highlights. EclipsePhoto should detect clipping and adjust exposure by 1 stop for the next frame.

16.3 Storage and Battery Margins

Eclipse photography generates a large volume of data in a short time, especially during the HDR bracket sequences of totality. The following table estimates storage requirements per rig:

SD Card Pre-Flight Verification. Before C1, EclipsePhoto performs automated storage pre-flight checks on each node:

- **Capacity check:** Verify at least 5× the estimated storage requirement is available (i.e., ≥65 GB free for a 13 GB estimated shoot). If below threshold, display a warning on the dashboard with the estimated number of brackets that can fit.
- **Write speed test:** Write a 500 MB test file to the SD card and measure throughput. UHS-II cards should sustain ≥80 MB/s; UHS-I cards at ~30 MB/s may cause buffer backlog during rapid bracket sequences. Flag cards below 50 MB/s as potentially problematic.
- **Card health:** Check for filesystem errors (`fsck` equivalent) and SMART-like wear indicators if the card supports them. A card with pre-existing errors should be replaced.
- **Test write/read/delete cycle:** Write a test frame, read it back, verify integrity (checksum), and delete it. This catches cards with read-only lock switches engaged or write-protected sectors.

SD Card Failure During Totality

An SD card that fills up or fails during totality cannot be swapped in time. The 2–7 minute window is too short to eject a card, insert a new one, wait for the camera to initialise it, and resume capture. Always use cards with at least 5× margin and verify write speed *before* C1.

Item	Conservative	Aggressive	Notes
Totality brackets	4.8 GB (96 frames)	9.6 GB (192 frames)	Continuous vs. 8 sets
Pre/post ramp brackets	—	11.2 GB (224 frames)	C2±5 min HDR timelapse
Partial phase captures	3.0 GB (60 frames)	10.0 GB (200 frames)	1/2min vs. 1/30s
C2/C3 bursts	5.0 GB (100 frames)	10.0 GB (200 frames)	50 vs. 100 per transition
4K video (slot 2)	—	8–15 GB	Continuous 10 min @ 100 Mbps
Total per rig	**~13 GB**	**~48 GB**	
Recommended card	**64 GB**	**128 GB UHS-II**	2.5× margin

Table 8. Storage requirements per rig: conservative vs. aggressive capture. The aggressive strategy fills a 128 GB card and produces enough data for HDR timelapse, video stacking, and temporal analysis.

Battery margins must account for the aggressive capture strategy: ~1,000 frames per rig requires at least two batteries (or a battery grip with dual cells). A typical mirrorless camera battery provides 400–500 shots (CIPA rating); the aggressive eclipse sequence demands 800–1,000 shots per rig. Use a battery grip, carry at least one spare per rig, and verify charge levels at T−4h.

16.4 Thermal Management

Cameras and Raspberry Pi nodes operating in direct sunlight can reach temperatures exceeding 45°C, especially in the 60–90 minutes of partial phase before totality. At these temperatures, sensor noise increases, the Raspberry Pi may throttle its CPU, and some cameras will display overheating warnings or shut down entirely.

Mitigation strategies include:

- **Reflective covers:** Use white or silver reflective material (space blankets, white foam board) to shade the camera body and Pi enclosure from direct sunlight.
- **Minimise live view:** Keep the camera's rear LCD off when not actively composing. Live view significantly increases sensor heating.
- **Monitor CPU temperature:** EclipsePhoto should read the Pi's thermal zone and warn if the CPU temperature exceeds 70°C (throttling typically begins at 80°C).
- **Shade structures:** If possible, set up a canopy or umbrella over the equipment. This also helps with screen visibility in bright sunlight.

Thermal Shutdown Risk

A camera that overheats and shuts down during totality cannot be recovered in time. The 2–7 minute totality window is too short to cool the camera, restart it, and reconfigure exposure settings. Thermal management must be addressed *before* C1, not during the event. Test the full equipment stack in direct sunlight for at least 90 minutes during the dry run.

17 Failure Mode Analysis

Eclipse photography is a zero-margin activity: any equipment failure during the 2–7 minute totality window results in permanent data loss. This section catalogs the most consequential failure modes, their probability, and EclipsePhoto's mitigation strategy.

Failure Mode	Severity	Prob.	Mitigation
Filter left on at C2	Critical	Medium	Audible alarm + full-screen warning + operator confirmation gate
Filter removed early (before C2)	Critical	Low	GPS-locked state machine prevents early FILTER_REMOVE transition
Camera overheat shutdown	High	Medium	Pre-eclipse shade, reflective covers, thermal monitoring with early warning
SD card full during totality	High	Low	Pre-flight storage check; 5× margin; auto-purge test frames
Focus drift (thermal)	Medium	High	Temperature monitoring; DOF margin at f/5.6+; pre-compensation
Wi-Fi link loss to hub	Medium	Medium	Each node runs autonomously; hub commands are optimization, not requirement
GPS time sync lost	High	Low	Fallback to NTP; manual C2 trigger button; pre-loaded ephemeris
Battery depletion mid-eclipse	High	Low	150% capacity margin; charge check at T−4h
Tracker stops / mount failure	High	Low	Wide rig on fixed tripod (no tracker); telephoto rigs have manual alt-az backup
Burst mode buffer full	Medium	Medium	Use fast UHS-II cards; limit burst to 20 frames; clear buffer between contacts

Table 9. Failure mode analysis for eclipse photography: severity and probability ratings with EclipsePhoto mitigation strategies. "Critical" means total loss of totality data; "High" means significant degradation.

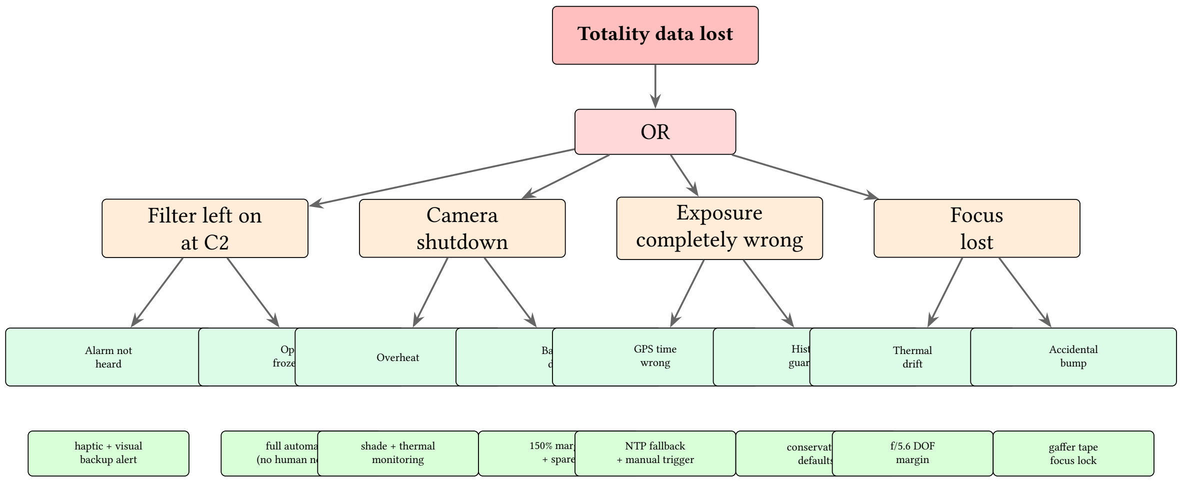


Figure 52. Fault tree for "totality data lost": the top-level failure decomposes into four primary causes, each with sub-causes and corresponding EclipsePhoto mitigations (green boxes). The system is designed so that no single point of failure can cause total data loss.

> **: No Single Point of Failure**
>
> Every critical function in the totality capture pipeline must have at least one fallback: GPS time has NTP backup, hub commands have autonomous node fallback, filter warnings have audio + visual + haptic channels, and bracket exposure has histogram guard rails. The dry run (Section 16) must verify each fallback path.

18 GPS Time Synchronization

Eclipse contact times are known to sub-second precision from published ephemeris data. To exploit this precision, EclipsePhoto's timing infrastructure must maintain accuracy within ±0.5 seconds of UTC. GPS provides this through its Pulse-Per-Second (PPS) signal.

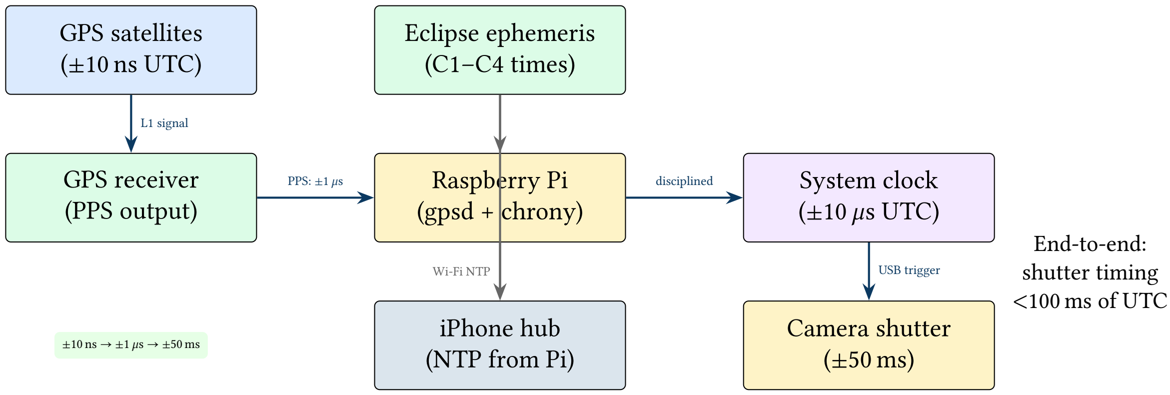


Figure 53. GPS time synchronization hierarchy: satellite signals provide nanosecond-accurate UTC, the GPS receiver's PPS output disciplines the Raspberry Pi's system clock via chrony, and USB camera triggers achieve <100 ms end-to-end timing accuracy. The iPhone hub synchronizes via NTP from the Pi nodes.

Each Raspberry Pi capture node includes a GPS module (e.g., u-blox NEO-M8N) connected via UART, with the PPS signal routed to a GPIO pin. The `gpsd` daemon reads NMEA sentences for coarse time, while `chrony` disciplines the system clock using the PPS edge for sub-microsecond accuracy.

The timing chain from GPS satellite to camera shutter involves several stages of degradation:

- **GPS to receiver:** ±10 ns (atomic clock accuracy, limited by ionospheric delay).
- **PPS to system clock:** ±1 μs (limited by GPIO interrupt latency on the Pi).
- **System clock to USB command:** ±5–20 ms (USB polling interval + protocol overhead).
- **USB command to shutter release:** ±30–50 ms (camera processing delay, model-dependent).

The total end-to-end accuracy of approximately ±50–100 ms is more than adequate for eclipse photography, where the fastest transient events (diamond ring) last 1–3 seconds.

GPS Cold Start

A GPS receiver requires 30–60 seconds for a cold start fix (up to 12 minutes in worst case without almanac data). Power on GPS modules at least 15 minutes before the eclipse. If operating in a canyon or under tree cover, consider an external antenna with clear sky view. Verify GPS lock status on the telemetry dashboard before C1.

: Time Source Fallback

If GPS lock is lost, fall back to NTP (accuracy ±50 ms over LAN from the iPhone hub, which has cellular NTP). If both GPS and NTP are unavailable, the operator can trigger C2/C3 transitions manually via the dashboard. The system must never silently use an unsynchronized clock—display a prominent warning if time confidence exceeds ±1 s.

19 Implementation Architecture

The preceding sections describe the astronomical science, optical engineering, and operational procedures that govern eclipse photography. This section maps those concepts to the EclipsePhoto codebase, providing a concrete bridge between the design rationale and its software realization.

19.1 Core Module Mapping

Table 10 maps each major report topic to the Python module that implements it, along with the principal classes and functions.

Table 10. Mapping of report sections to EclipsePhoto Python modules.

Report Concept	Python Module	Key Classes / Functions
Eclipse geometry & contacts (§2)	`core/ephemeris.py`	`EclipseEphemeris`, `ContactTimes`
Light levels & EV (§3)	`core/illuminance.py`	`IlluminanceModel`, `ev_for_phase()`
Stage detection (§4)	`core/stage_detector.py`	`StageDetector`, `Phase` enum
Exposure adaptation (§8)	`core/exposure.py`	`ExposureController`, `BracketSequence`
Solar filter management (§9)	`hardware/filter_controller.py`	`FilterController`, `FilterState`
Multi-rig coordination (§10)	`core/coordinator.py`	`RigCoordinator`, `NodeSync`
HDR pipeline (§11)	`core/hdr_pipeline.py`	`HDRMerge`, `LarsonSekanina`
State machine (§13)	`core/state_machine.py`	`EclipseStateMachine`, `State` enum
Focus detection (§15)	`core/focus.py`	`FocusAnalyzer`, `LaplacianVariance`
GPS sync (§18)	`hardware/gps.py`	`GPSTimeSource`, `PPSSync`
Planning app (§20)	`interface/planning.py`	`PlanningServer`, `SkyChart`

19.2 Hardware Abstraction

EclipsePhoto isolates hardware-specific details behind abstraction layers, allowing the core logic to remain camera- and platform-agnostic:

- **Camera control:** Primary camera communication uses `gphoto2` Python bindings for broad camera compatibility, with a dedicated Sony Remote SDK driver for Sony Alpha cameras that enables lower-latency shutter control and direct access to Sony-specific features (pixel-shift multi-shot, real-time tracking AF status).
- **GPIO interface:** Solar filter servo control and GPS PPS input are accessed through `RPi.GPIO` or `gpiozero` on Raspberry Pi nodes. The GPIO layer abstracts pin assignments via the YAML configuration file (see Section 19.4), allowing different Pi models and wiring layouts without code changes.
- **USB device management:** Multi-camera nodes may connect to more than one USB device (camera, GPS receiver, filter servo controller). The USB management layer uses `pyudev` to detect and map devices by serial number, ensuring stable device assignment across reboots.
- **Network layer:** Each Pi node exposes a Flask/FastAPI REST API for command reception and status reporting. Real-time telemetry (histogram data, shutter events, filter state transitions) is streamed to the iPhone companion app over WebSocket connections, providing low-latency updates for the operator dashboard (Section 14).

19.3 Data Flow Architecture

Figure 54 illustrates the real-time data flow through the EclipsePhoto capture pipeline. GPS-disciplined time feeds the ephemeris engine, which determines the current eclipse stage. The stage detector drives both the exposure controller (for bracket sequence selection) and the filter controller (for safety-critical filter state transitions). A histogram feedback loop from captured frames allows the exposure controller to make real-time adjustments within the bounds of the active bracket sequence.

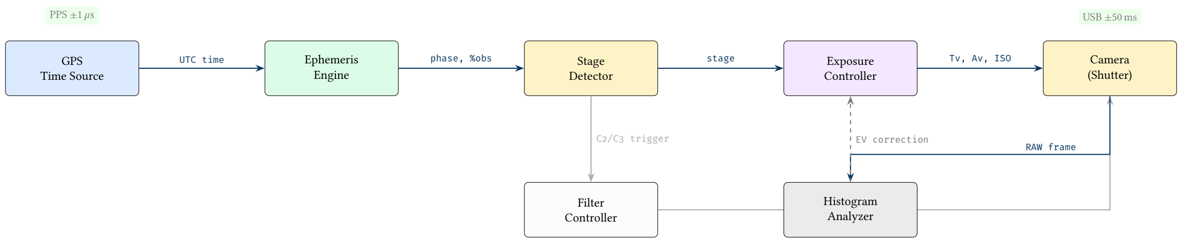


Figure 54. Real-time data flow through the EclipsePhoto capture pipeline. GPS-disciplined time feeds the ephemeris engine, which determines the eclipse stage. The stage detector drives both the exposure controller and the filter controller. Histogram analysis of captured frames provides feedback to the exposure controller for real-time EV adjustments. Orthogonal routing reflects the actual message-passing architecture between Python modules.

19.4 Configuration as Code

All eclipse parameters and per-rig settings are specified in a YAML configuration file, making each deployment fully reproducible and version-controlled:

- **Eclipse parameters:** Observer location (latitude, longitude, altitude), eclipse date, and contact times. Contact times may be specified explicitly (from published NASA data) or computed automatically from Besselian elements embedded in the ephemeris module.
- **Per-rig configuration:** Camera model, lens focal length, sensor size, filter type (glass ND5 vs. film), bracket sequence definition (number of exposures, EV spacing, base exposure per stage), and burst mode settings.
- **Design rule validation:** The configuration loader validates all settings against the design rules specified throughout this report. For example, it verifies that the configured bracket sequence spans at least 12 EV during totality (Section 8), that the filter removal window falls within the safe C2 margin (Section 9), and that the GPS cold-start allowance is at least 15 minutes (Section 18).
- **Startup checks:** At system startup, EclipsePhoto loads the YAML configuration, validates all constraints, and prints a summary of any violations. The system refuses to enter autonomous mode if critical constraints are violated (e.g., no bracket sequence defined for totality, no filter removal time configured).

19.5 Sony Alpha Camera Integration

EclipsePhoto v1 targets Sony Alpha mirrorless cameras—including the A7 series (A7R V, A7 IV, A7S III), the A1, and the A9 III—as its primary capture hardware. Sony Alpha cameras are selected for their combination of high-resolution sensors, excellent dynamic range, silent electronic shutter, and robust USB remote control capabilities.

Key capabilities used by EclipsePhoto:

- **Remote shutter and exposure control:** Full programmatic control of shutter speed, aperture, ISO, and drive mode via USB, enabling the automated bracket sequences described in Section 8.
- **Burst mode and continuous shooting:** High-speed burst modes (up to 30 fps on the A1, 120 fps electronic shutter on the A9 III) support rapid bracket capture during the brief diamond ring and chromosphere flash transients.
- **RAW capture:** 14-bit uncompressed or losslessly compressed RAW files provide the bit depth and dynamic range required for HDR compositing (Section 11).
- **Video capabilities:** The A7S III and A7 IV support 4K at 60 fps; the A1 supports 4K at 120 fps. These high-frame-rate video modes are useful for video stacking (extracting and combining individual frames, similar to lucky imaging in planetary photography) during the diamond ring and Baily's Beads phases where atmospheric seeing limits single-frame sharpness.
- **RAW video profiles:** S-Log3 and HLG gamma curves provide maximum dynamic range in video recording, preserving highlight and shadow detail for post-processing. These LOG profiles are particularly valuable during the diamond ring, where the brightness ratio between the last sliver of photosphere and the inner corona exceeds 10 EV.
- **Simultaneous stills and video:** Some Sony models support proxy video recording concurrent with RAW still capture (with limitations on resolution and frame rate), enabling a belt-and-suspenders approach to diamond ring coverage.

19.6 Future Camera Support

The hardware abstraction layer described above is designed for extensibility. While v1 targets Sony Alpha exclusively, the architecture accommodates additional camera platforms:

- **Canon EOS R series:** Canon's Camera Control API (CCAPI) provides HTTP-based remote control for EOS R mirrorless cameras, which maps cleanly onto EclipsePhoto's REST-based node architecture.
- **Nikon Z series:** The Nikon SDK offers USB and wireless control for Z-mount mirrorless cameras. Nikon's strong high-ISO performance (particularly on the Z8 and Z9) makes these cameras attractive for deep corona imaging.
- **Generic gphoto2-supported cameras:** The existing `gphoto2` driver layer provides baseline compatibility with over 2,500 camera models. While advanced features (burst timing, LOG video) may not be available through `gphoto2`, basic remote shutter and exposure control is sufficient for a supplementary rig.
- **Smartphone cameras as auxiliary sensors:** Modern smartphone computational photography pipelines (multi-frame stacking, HDR fusion, night mode) can produce surprisingly capable eclipse images. A smartphone mounted alongside the primary rig can serve as an auxiliary sensor, contributing wide-field context frames via its ultra-wide lens.

- **Dedicated astro cameras:** Cooled CMOS astro cameras (ZWO ASI series, QHY series) offer high-frame-rate narrowband imaging capabilities relevant to prominence and chromosphere capture. These cameras connect via USB 3.0 and provide 12- or 16-bit data at hundreds of frames per second in region-of-interest modes, enabling lucky-imaging workflows during totality.

20 Eclipse Planning and Companion App

The iPhone companion app serves a dual role in EclipsePhoto: during the eclipse it acts as the hub for real-time telemetry and operator commands (Sections 14–16), but in the days and weeks before the event it functions as a comprehensive planning tool. Pre-eclipse preparation—selecting a site, verifying equipment configurations, rehearsing the timeline—is where most eclipse photography failures are prevented. This section details the planning features that the companion app provides.

20.1 Eclipse Circumstance Calculator

The app computes contact times (C1, C2, C3, C4) for any observer location using Besselian elements published by the International Astronomical Union and NASA. Given the observer's GPS-derived latitude, longitude, and altitude, the calculator determines:

- **Contact times** for C1 (first contact, partial eclipse begins), C2 (second contact, totality begins), C3 (third contact, totality ends), and C4 (fourth contact, partial eclipse ends), each computed to ±1 s precision using the device's GPS-derived position.
- **Countdown timers** to each contact point, updating in real time.
- **Duration of totality** (C3 − C2) at the selected location.
- **Eclipse magnitude** (fraction of the solar diameter obscured at maximum eclipse) and **obscuration percentage** (fraction of the solar disc area covered).
- **Solar altitude and azimuth** at each contact point, critical for site selection (avoid obstructions along the solar bearing) and for estimating atmospheric dimming.
- **Path width and centre-line offset**: the distance from the selected location to the centre line of the path of totality, and the local path width.

> **: GPS Position for Contact Times**
>
> Contact time accuracy is dominated by position accuracy, not ephemeris precision. A 1 km error in position can shift contact times by up to 2 seconds. The app requires a GPS fix with horizontal accuracy better than 10 m before displaying contact times, and continuously refines the prediction as the fix improves.

20.2 Sky Chart and Visible Object Prediction

During totality, the sky darkens sufficiently to reveal planets and bright stars in the vicinity of the Sun (as discussed in Section 4). The companion app provides an interactive sky chart centred on the Sun's position at mid-totality, showing surrounding celestial objects that may be visible or captured in wide-angle photographs.

Objects are colour-coded by predicted visibility:

- Definitely visible (bright planets: Venus, Jupiter)—these will appear prominently in images and are useful as focus-confirmation targets.
- Probably visible (Mars, Mercury, bright stars such as Sirius or Regulus)—visible under good sky transparency conditions.
- Possibly visible (fainter stars, magnitude ~2–3)—depends on sky transparency, altitude above horizon, and totality duration.

The sky chart is equally valuable *after* the eclipse: when reviewing wide-angle images, that unidentified dot 10° from the eclipsed Sun can be cross-referenced against the predicted object map. Pre-computing object positions eliminates guesswork during post-processing.

Insight: *Post-Processing Identification*

20.3 Solar Activity Monitor

The visual character of the corona and prominences depends on the Sun's activity level at the time of the eclipse. The companion app integrates real-time data from NOAA's Space Weather Prediction Center (SWPC) to inform framing and exposure decisions:

- **Solar flare alerts:** active region tracking with X-ray flux monitoring from GOES satellites.
- **Active region map:** locations of sunspot groups on the solar disc, with rotation projections to estimate which regions will be near the limb on eclipse day.

- **Prominence monitoring:** SDO/AIA 304 Å imagery showing hydrogen-alpha prominences on the solar limb, updated every 12 minutes. This reveals which side of the Sun currently hosts large prominences.

> **: Framing Bias from Prominence Position**
>
> If SDO imagery shows a large active region or prominence approaching the west limb in the days before the eclipse, the telephoto rigs should frame with additional margin on that side to capture the prominence during totality. The app can suggest a framing offset based on current prominence positions and solar rotation rate (~13.2°/day at the equator).

20.4 Weather and Site Selection

Cloud cover is the single greatest threat to eclipse observation. The companion app aggregates weather data to support site selection and day-of escape decisions:

- **Cloud cover forecasts** for candidate viewing sites along the path of totality, drawn from multiple numerical weather prediction models (GFS, ECMWF, NAM).
- **Satellite imagery overlay:** GOES visible and infrared channels displayed on a map with the eclipse path, updated every 10–15 minutes on eclipse day.
- **Clear sky probability scoring:** a composite score for each candidate site based on historical cloud climatology, current forecast models, and satellite trends. Sites are ranked from best to worst.
- **Escape route suggestions:** if clouds threaten the primary site on eclipse morning, the app suggests alternative sites along the path reachable by car within the available time, ranked by clear sky probability.
- **Wind forecast:** surface wind speed and direction, relevant both for shadow band visibility (shadow bands are more visible in calm conditions) and for tripod stability (gusts above 30 km/h can introduce vibration in long telephoto setups).

> **Weather Decision Deadline**
>
> The decision to relocate must be made early—typically 2–4 hours before C1—to allow time for travel, setup, GPS lock acquisition, and a full dry run at the new site. Waiting for "one more forecast update" is a common trap that leaves the operator scrambling.

20.5 Long-Range Site Selection and Climate Analysis

While the weather subsection above addresses short-range forecasting and day-of decisions, site selection should begin 12–24 months before the eclipse. The choice of primary observing site is the single highest-leverage decision in the entire project—no amount of automation or optical engineering can compensate for clouds.

20.5.1 Climatological Cloud Cover

Historical satellite data from GOES and MODIS archives provide cloud-cover statistics for any location on earth, resolved by month and even by time of day. For the eclipse date ±2 weeks across 10–20 years of archived imagery, one can compute a clear-sky probability for each candidate site. Locations along the same eclipse path can have dramatically different cloud statistics: coastal sites may suffer persistent marine layer, while sites 100 km inland on the leeward side of a mountain range may enjoy reliably clear skies.

20.5.2 Rainfall and Monsoon Patterns

The eclipse date must be checked against seasonal precipitation patterns for each candidate site. The 2026 August 12 eclipse passes through regions with highly variable summer weather: Iceland in August averages roughly 50% cloud cover with frequent Atlantic weather systems, while northeastern Spain (Aragon, Catalonia) benefits from a Mediterranean summer climate with substantially lower cloud probability. Monsoon-affected regions, tropical wet seasons, and locations prone to afternoon convective buildup should be identified and deprioritized.

20.5.3 Time-of-Year and Altitude Considerations

The season of the eclipse affects both sky conditions and operational logistics. Winter eclipses offer cold, dry air that can produce excellent transparency, but introduce equipment challenges (battery capacity drops 10–30% in freezing temperatures, dew/frost risk, operator discomfort). Summer eclipses in desert or semi-arid regions (e.g., the American Southwest for the 2045 eclipse) combine warm conditions with excellent seeing.

Higher elevation sites (2,000 m and above) reduce the atmospheric column between the camera and the corona, improving both transparency and seeing. Mountain observatories are ideal if accessible. Even a modest altitude gain of 500–1,000 m above the surrounding terrain can meaningfully reduce atmospheric dimming, particularly for the faint outer corona.

20.5.4 Atmospheric Seeing and Airflow

For telephoto corona imaging at 500–800 mm, atmospheric seeing directly limits achievable resolution. Sites with laminar airflow—coastal locations with steady onshore winds, high plateaus, or locations downstream of smooth terrain—produce steadier seeing than sites surrounded by thermally active surfaces (asphalt, rocky desert in afternoon sun). Seeing also affects the visibility of shadow bands, the rippling patterns visible on flat surfaces in the final minute before and after totality.

20.5.5 Accessibility and Infrastructure

Clear-sky probability must be balanced against logistics. A remote desert site with 95% clear-sky odds but no road access, no accommodation, and no mobile network is operationally worse than a town with 85% odds and full infrastructure. Key considerations include:

- Road quality and travel time from the nearest airport or accommodation.
- Mobile network coverage for real-time weather updates and remote telemetry.
- Availability of mains power or the need for battery/generator operation.
- Security of equipment left overnight at a remote site.
- Local regulations, permits, or restricted areas along the eclipse path.

20.5.6 Contingency Planning

No single-site plan is robust against weather. The operator should identify 2–3 fallback sites along the eclipse path, each separated by at least 100 km, so that fundamentally different weather systems affect each site. If the primary site clouds over on eclipse morning, there must be time to relocate to an alternate site and complete a full setup and dry run before C1.

Table 11. Climate comparison for candidate 2026 August 12 eclipse sites.

Site	Avg. cloud cover (Aug)	Rain probability	Altitude	Access
NW Iceland	55–65%	Moderate	Sea level	Good (Ring Road)
NE Spain (Aragon)	20–30%	Low	500–1,000 m	Excellent
Ibiza / Mallorca	15–25%	Very low	Sea level	Excellent

: Site Selection Timeline

Begin site selection 12–24 months before the eclipse. By 6 months out, the primary site and at least two alternates should be identified, with accommodation and logistics arranged. Climate data, not convenience, should drive the decision.

20.6 Equipment Configuration Wizard

The equipment configuration wizard guides the operator through systematic pre-eclipse setup:

- **Rig definition:** select cameras, lenses, and sensor sizes for each capture node. The app computes the resulting field of view and solar disc size in pixels for each rig.
- **Framing preview:** a simulated view through each rig's optics showing the eclipsed Sun, inner and outer corona extent, and surrounding bright objects, scaled to the sensor's pixel dimensions.
- **Bracket sequence builder:** configure HDR exposure ranges for each eclipse phase (partial filtered, diamond ring, inner corona, outer corona, prominences). The builder displays the resulting EV coverage and warns if gaps exist in the bracket range.
- **Storage calculator:** given the configured shot count per phase, file format (RAW, compressed RAW, JPEG), and file size estimates, the calculator projects total storage consumption against the inserted card capacity. A warning is issued if projected usage exceeds 80% of card capacity.
- **Battery life estimator:** based on shot count, continuous live-view usage, ambient temperature (cold temperatures reduce battery capacity by 10–30%), and Wi-Fi telemetry overhead, the estimator projects remaining battery life through C4.

Both storage and battery estimates include a 50% safety margin: the system budgets as though totality will last 1.5× the predicted duration, because the operator may extend the bracket sequence if conditions warrant, and because partial-phase monitoring consumes resources for over an hour before totality begins.

Insight:
Storage and Battery Margins

20.7 Rehearsal and Dry Run Mode

The rehearsal mode is arguably the single most important planning feature. It simulates the full eclipse timeline at accelerated speed (10×, 50×, or 100× real-time), exercising the complete automation pipeline:

- Each Pi node executes its full capture sequence, issuing actual camera commands (if a camera is connected) or logging simulated commands (if no camera is connected).
- The operator monitors the telemetry dashboard exactly as they would during the real event, verifying that filter removal/replacement timing is correct and that bracket sequences fire as expected.
- Solar filter state transitions are verified: the system confirms that filter removal occurs within the correct window before C2 and that filter replacement occurs before C3 egress.
- After each dry run, the app presents a detailed log review: a timestamped sequence of every command, state transition, and telemetry event, colour-coded to highlight any sequencing errors or timing violations.
- Historical replay is supported: the operator can load the ephemeris data from a previous eclipse (e.g., the April 2024 eclipse over North America) and run the full sequence against those contact times, enabling realistic practice months before the target event.

: Mandatory Dry Run

At least one full dry run at the deployment site—with all cameras connected and solar filters physically cycled—must be completed before C1. The dry run log must show zero sequencing errors before the system is considered “go” for the real event. This is the single most effective measure against automation failures during totality.

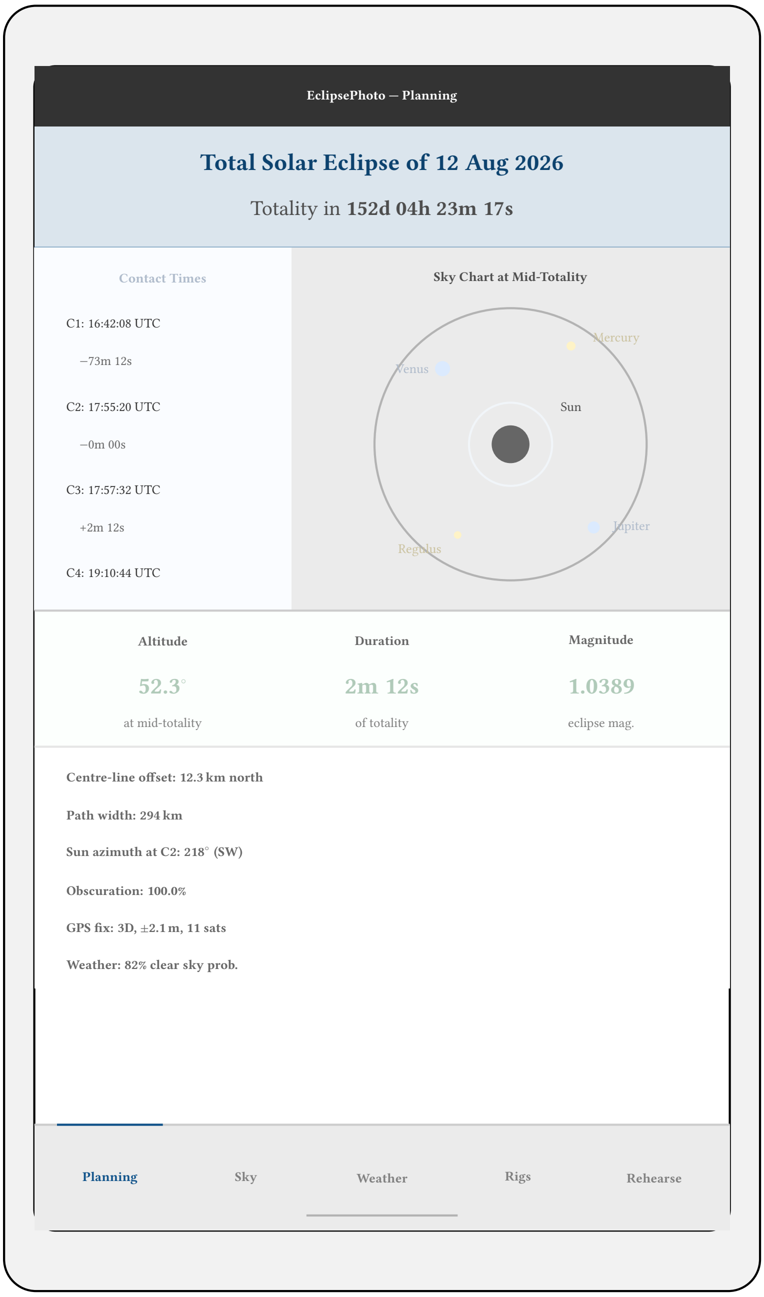


Figure 55. Companion app planning screen mockup. The top header shows the eclipse date and a countdown timer. The left panel lists contact times (C1–C4) with countdowns relative to C2. The centre shows an interactive sky chart with colour-coded celestial objects (green: definitely visible; orange: probably visible; red: possibly visible). The bottom section displays quick statistics and site data. Five tab icons along the bottom provide access to the planning, sky chart, weather, rig configuration, and rehearsal modules.

Revision History

Version	Date	Changes
0.1.0	March 2026	Initial technical report; eclipse science + operational design